

TEL AVIV UNIVERSITY

The Iby and Aladar Fleischman Faculty of Engineering

The Zandman-Slaner School of Graduate Studies

**MAPPING LOSSES THROUGH EMPIRICAL
EXTRACTION OF THE SPATIAL EXTERNAL
LUMINESCENCE EFFICIENCY**

A thesis submitted toward the degree of
Ph.D. in Electrical Engineering

by

Tamir Yeshurun

February 2026

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This research was carried out at Tel Aviv University

in the School of Electrical Engineering

Faculty of Engineering

under the supervision of Prof. Gideon Segev

February 2026

*This work is intended to promote our sustainable future ever so slightly,
and is therefore dedicated to all.*

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Abstract

With the escalating climate crisis that is significantly driven by our reliance on fossil fuels, more sustainable energy sources are increasingly required. With anticipated increases in electricity demand due to globalization, expanding access to energy, and a growing world population, the need for robust and scalable renewable solutions has never been more critical. In recent years, Photovoltaic (PV) cells have emerged as the leading candidate to meet this challenge.

The predominant solar cell technology, that is based on silicon (Si) single-junction cells, offers reliable and long-lasting conversion of solar radiation conversion into electricity. These devices have achieved impressive efficiencies, with records reaching approximately $\approx 27.6\%$, nearing the physical thermodynamic limit of 33% for this material. [1] Higher efficiencies, up to 47.6%, have been demonstrated with alternative materials and designs. Notably, silicon single-junction PV cells have recently become the most cost-effective method for energy production, surpassing both conventional and other renewable sources.¹ Nevertheless, any advance in PV energy conversion technologies and efficiency will directly translate to reduced land requirements for energy generation, increased production and distribution volumes, and ultimately, lower costs and broader adoption. Therefore, enhancing the energy conversion efficiencies of solar cells, is crucial for securing a sustainable energy future.

The efficiency of PV cells is determined by charge transport properties and loss mechanisms within the device. Quantitative assessment of these mechanisms is fundamental for device optimization and the evaluation of new materials and interfaces. This work introduces a method for depth-resolved quantitative mapping of loss and charge transport mechanisms under operando conditions. By coupling photoluminescence (PL) measurements together with optical modeling, we introduce and demonstrate the extraction of the spatial external luminescence efficiency (SELE) depth profile. The SELE profile is defined as the probability for a photo-generated electron-hole pair at a specific depth within a device, to contribute to its PL emission. Thus, mapping the competition between radiative and non-radiative recombination within the device and can thus be used to study the properties of surfaces and interfaces. Since the external luminescence efficiency is related to the obtainable photo-voltage from the device, the SELE can be used to map the spatial contribution to the device photo-voltage.

¹ Lazard, Levelized Cost of Energy Analysis, Version 17.0, 2024

The SELE extraction method is established using model GaAs wafers from both measurements and COMSOL simulations. PL simulations are shown alongside a photon recycling model. These simulations validate the extraction method and attribute features in the profile to important physical quantities such as the surface recombination velocity and carrier lifetime. The development of the method follows by an investigation of the SELE of PN junctions and the effects of different surface properties on the SELE profile of MOS capacitors using simulations. Lastly, the dynamic PL measurements SELE of native oxide layer formation are shown.

Nomenclature

ELE External Luminescence Efficiency

EQE External Quantum Efficiency

GaAs Gallium Arsenide

InP Indium Phosphide

PGD Projected Gradient Descent

PL Photoluminescence

PV Photovoltaic

SCE Spatial Collection Efficiency

SELE Spatial External Luminescence Efficiency

Si Silicon

TAU Tel-Aviv University

TIR Total Internal Reflection

TMM Transfer Matrix Method

TOF-SIMS Time of Flight Secondary Ions Mass Spectroscopy

USC Ultrasonic Cleaner

VASE Variable Angle Spectroscopic Ellipsometer

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1 Introduction

1.1 Overview

Photovoltaic (PV) technology plays a critical role in the global transition toward sustainable energy by directly converting sunlight into electricity. Devices are commonly measured by their power conversion efficiency, which quantifies the ratio of incident solar energy that is converted into electrical power. The performance and energy conversion efficiency of PV devices is fundamentally determined by the competition between charge carrier collection and the various recombination processes throughout the device.[3] Absorption of photons with energy above the material band-gap, excites mobile charge carriers within the device. Generated charge carriers can diffuse, drift, interact via different recombination processes, and be collected as current. Charge carriers are then able to transfer the available energy as output electrical energy. Any factor for loss of carrier density can therefore indicate a potential loss to the device performance.

Recombination processes are thus the main factor in loss of charge carrier density and affects the energy conversion efficiency of the device. These recombination processes can either be radiative, where an excited electron interacts with a nearby hole, which results in a release of energy in the form of a photon. The other types of recombination are non-radiative recombination, where the interaction between the charge carriers results in energy that is dissipated via phonons. Perfect solar energy conversion devices experience no non-radiative recombination, so that more generated charge carriers reach the contacts. However, device structure, material properties, charge mobility, lifetime, and optical design all affect performance. At any point in the device, there exists competition between different recombination mechanisms and charge collection. The study and characterization of these potential loss mechanisms can help develop methods for improving the performance of PV devices.

This work presents the study of spatial external luminescence efficiency (SELE) as

a new characterization method for PV devices. Various characterization methods have proven valuable in assessing the overall performance and material quality of PV devices, yet they often provide limited spatial information on defects and recombination processes within the device. While techniques like EQE and PL-imaging can offer insights into charge collection from different areas of the device, they do not fully map the device across its different regions and interfaces. To address these limitations, this work introduces the SELE analysis. The SELE provides a spatially resolved analysis of the external luminescence efficiency (ELE) using PL measurements. This method reveals how material properties and device design influence light emission. SELE serves as a new tool for optimizing PV device performance by identifying specific areas of potential energy loss.

1.2 Characterization methods for PV Devices

To evaluate PV device performances, various characterization methods are available. Some methods focus on the applicative performance of the entire cell, usually by observing the output current under different conditions. These include the current-voltage (I-V) measurement, that is usually performed under the illumination standard of "1 sun" conditions. This method encapsulates different aspects, properties, and variables into one simple result, which corresponds to the output power obtainable from the device. However, in this method, the output current is the net effect of all internal processes. Thus, this method does not distinguish the effects of each specific loss mechanisms throughout the device.

A complementary measurement to that of the I-V includes the External Quantum Efficiency (EQE) measurement. In this method, the output current of a device is measured for a range of monochromatic illumination. EQE offers insight into wavelength-dependent performance and can help locate parasitic absorption or optical losses. However, it does not provide direct information on recombination dynamics or voltage losses. Moreover, both I-V and EQE measurements require fully fabricated devices with electrical contacts, limiting their use during early stages of material or device development.

Other methods investigate surface or bulk properties. The surface properties of PV devices play a critical role in determining the overall efficiency of the device. Surface defects, such as dangling bonds or contamination, can introduce electronic states within the band-gap that act as recombination centers for charge carriers. This non-radiative recombination reduces the number of carriers collected as current, directly lowering the short-circuit current, open circuit voltage and overall efficiency. Effective surface characterization methods are therefore crucial to understand recombination losses at the surface.

Surface photovoltage (SPV) is one technique used to study the electronic properties of semiconductor surfaces. In this measurement, the change in surface potential is measured under illumination, where photo-generated electron-hole pairs are separated by built-in electric fields near the surface, leading to a redistribution of charge and a measurable change in surface potential. The magnitude and dynamics of this photovoltage response provide insight into some key surface properties. However, this method is limited by nature to the changes at the surface only and cannot be implemented to interfaces in a layered device structure. Moreover, this method does not include the influence of various bulk dynamics and properties.

Bulk properties are also fundamental to device performance as they determine how efficiently photo-generated charge carriers can move through the material and reach the collecting junctions without recombining. Several techniques are available to characterize the bulk properties of photovoltaic materials, each offering insight into different aspects of carrier dynamics and material quality. Such methods include photo-conductance decay, a measurement used to evaluate the effective carrier lifetime, and Hall effect measurements which provide information on bulk carrier mobility. However, these methods are typically sensitive to the thickness of the entire device, making them suitable for assessing the effective bulk lifetime but not for depth-resolved analysis.

1.3 Photoluminescence (PL)

Semiconductor materials can absorb light, generating mobile charge carriers. Conversely, an opposite process occurs when charge carriers recombine radiatively, which results in the emission of a photon with an energy that corresponds to the material band-gap. This process occurs spontaneously and is thus named the spontaneous radiative recombination. Light emission from an excited substance is termed luminescence, and light emission due to external photoexcitation is referred to as photoluminescence (PL). Many factors influence carrier radiative recombination. Local non-radiative recombination centers, such as surface recombination or SRH recombination, can influence the amount of carriers that undergo radiative recombination locally. For example, non-passivated surfaces can induce a high surface recombination velocity, which results in lower photoluminescence near the surface. Thus, the luminescence from a device carries information about local recombination processes and loss mechanisms. For this reason, PL measurements are used to study different device properties, such as band-gap energy, the presence of impurities or defects, carrier lifetime, and internal quantum efficiency.

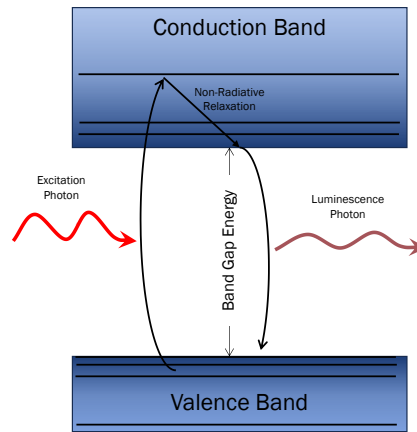


Figure 1.1: Schematic illustration of the photoluminescence process in a semiconductor. Incident illumination with an energy higher than the band gap energy of the semiconductor excites electrons from the valence band to the conduction band, creating pairs of electron and holes. As the electrons recombine with holes photons are emitted with energy corresponding to the material's band gap. The luminescence that escape the material is termed photoluminescence.

PL measurement is a common characterization method for photovoltaic devices which offers non-destructive insights into material properties and performance. Similar methods such as time-resolved photoluminescence (TRPL) and hyperspectral imaging enable the assessment of carrier dynamics and defect states and allow the quantification of losses due to trapping defects. [4] While lateral spatially-resolved PL mapping can also be used to provide information on sub-cell performance and manufacturing inconsistencies without damaging samples. [5] The combination of steady-state and modulated PL techniques further enhances the understanding of recombination processes and carrier lifetimes, which are crucial for optimizing device efficiency. These PL techniques enable a in-depth and rapid characterization of the optoelectronic properties, accelerating the development, and identifying performance bottlenecks.

1.4 Spatial Collection Efficiency (SCE)

The performance of solar energy conversion devices depends on the relationship between charge carrier collection and various recombination processes within the device. Photo-generated charge carriers either recombine through non-radiative or radiative processes or can be collected and contribute to the output current. The probability for charge carriers to recombine or contribute to current varies significantly within the device. The SCE, $S_{CE}(z)$ is defined as the probability that photo-generated electron hole pairs that are generated at some point z within a device contribute to the collected current density at the contacts, J_{out} . Therefore, the output current density that is induced by illumination with a monochromatic incident beam at wavelength λ_{in} follows:

$$J_{out}(\lambda_{in}) = q \cdot \int_0^d G(\lambda_{in}, z) S_{CE}(z) dz \quad (1.1)$$

Where $G(\lambda_{in}, z)$ is the optical generation that is induced by the monochromatic light, and d is the device thickness. The external quantum efficiency (EQE), is defined as the fraction of incident photons that are collected as current:

$$EQE(\lambda_{in}) = \frac{1}{q} \frac{J_{out}(\lambda_{in})}{\phi_{inc}(\lambda_{in})} \quad (1.2)$$

where $\phi_{inc}(\lambda_{in})$ is the incident monochromatic photon flux. Combining equations (1.1) and (1.2) yields:

$$EQE(\lambda_{in}) = \frac{1}{\phi_{inc}(\lambda_{in})} \int_0^d G(\lambda_{in}, z) S_{CE}(z) dz \quad (1.3)$$

Previous works have shown that a careful examination of the spatial collection efficiency (SCE) can be used to map losses in photo-voltaic (PV) and photo-electrochemical (PEC) cells [6]. The SCE provides a functional depth profile of the active regions in the device under *operando* conditions. Analysis of SCE profiles allowed estimating diffusion and collection lengths, differentiating between bulk and surface losses, and quantifying the probability that absorbed photons do not generate mobile charge carriers in metal oxides.[7] Although the SCE provides extensive information on current collection, it cannot be used to characterize surface passivation layers, which are essential for reducing surface recombination losses, nor provide information on the contribution to the cell voltage. Hence, a new complementary characterization method that can map and quantify these properties is called for.

1.5 External Luminescence Efficiency (ELE)

Under standard operating conditions, photo-generated electron-hole pairs that are not collected as current, recombine through bulk and surface defect states or through radiative recombination, thus generating photons. These photons are either reabsorbed within the device (i.e. photon recycling) or emitted through photoluminescence (PL). The fraction of the photogenerated electron hole pairs that recombine radiatively is the internal luminescence yield, and the probability for any recombination event to result in photon emission is the external luminescence efficiency, η_{ext} (ELE).[8, 9, 10, 11] Since both the radiative recombination rate and the photo-voltage are determined by the separation of the quasi-Fermi levels for electrons and holes, the open circuit voltage that can be obtained from a light absorber, is tightly coupled to its ELE:[10, 11, 12]

$$qV_{oc} = qV_{oc,ideal} + kT \ln(\eta_{ext}) \quad (1.4)$$

Where $V_{oc,ideal}$ is the open circuit voltage of an ideal light absorber, k is the Boltzmann constant, q is the fundamental charge, and T is the device temperature. Thus, ELE measurements provide an estimate of the maximum open circuit voltage that can be obtained from a light-absorbing material.[11] Consequently, device optimization towards maximizing the extraction of radiative recombination photons led to a significant increase in the performance of GaAs solar cells, achieving a record efficiency of 28.8%. [11];

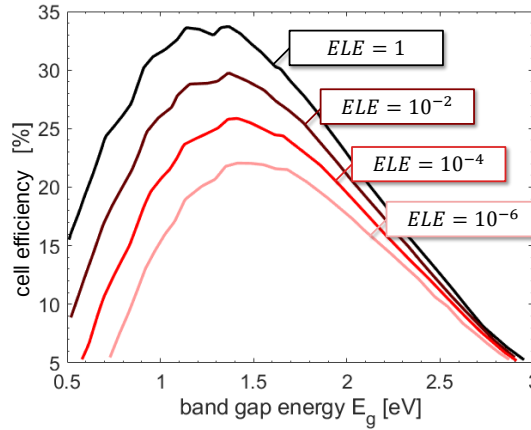


Figure 1.2: Single junction solar cell maximum conversion efficiency with respect to the material's band gap energy, for different values of ELE (η_{ext}). [13]

Various approaches were developed to study important device parameters using PL measurements. Voltage-dependent PL imaging was used to expose non-ideal charge ex-

traction regions [14, 15]. A lateral map of the quasi-Fermi-level separation averaged in depth was obtained using hyperspectral PL imaging [16]. Lombez et al. showed that combining voltage-dependent luminescence measurements with electrical transport simulations can provide a qualitative depth analysis of device properties [17]. Tennyson et al. produced tomographic photo-voltage images of CIGS solar cells by acquiring photo-voltage maps of devices illuminated at different wavelengths [18]. However, since there was no attempt to deconvolute the correlation between the optical generation depth profile and the obtained photo-voltage, only qualitative depth information was obtained. Despite providing a wealth of information on device performance and material properties, these methods do not map losses *quantitatively* along the depth profile of the device.

1.6 Spatial External Luminescence Efficiency (SELE)

The spatial external luminescence efficiency is defined as the probability that photo-generated electron hole pairs that are generated at some point z within a device contribute to the measured PL photon flux, ϕ_{out} . Therefore, the total PL photon flux induced by illumination with a monochromatic incident beam at wavelength λ_{in} follows:

$$\phi_{out}(\lambda_{in}) = \int_0^d G(\lambda_{in}, z) S(z) dz \quad (1.5)$$

As in equation (1.1), $G(\lambda_{in}, z)$ is the optical generation that is induced by the monochromatic light, and d is the device thickness. The SELE is the correlation between the photoexcitation position in which charge-transport processes commence, and the eventual contribution to photoemission after undergoing transport processes. Thus, SELE profiles account for the effects of drift and diffusion, all recombination mechanisms, surfaces, and junction formations. Furthermore, SELE profiles extracted from equation (1.5) account for these mechanisms with no assumptions on the charge transport mechanisms, position of junctions or the effects of surfaces.

Under open circuit conditions, the ELE is the fraction of absorbed photons that are emitted in the PL process:

$$\eta_{ext}(\lambda_{in}) = \frac{\phi_{out}(\lambda_{in})}{\phi_{abs}(\lambda_{in})} \quad (1.6)$$

Where ϕ_{abs} is the flux of photons that are absorbed in the sample. Combining equations (1.5) and (1.6) yields:

$$\eta_{ext}(\lambda_{in}) = \frac{1}{\phi_{abs}(\lambda_{in})} \int_0^d G(\lambda_{in}, z) S(z) dz \quad (1.7)$$

Thus, combining optical modeling to calculate $G(\lambda_{in}, z)$, with measurements of $\eta_{ext}(\lambda_{in})$, and $\phi_{abs}(\lambda_{in})$, the SELE profile can be extracted from equation (1.7) using various regularization tools.

Using device simulation, the SELE can be calculated directly by finding the change in the PL photon flux that results from a small perturbation in optical generation at a specific position within the device. For a one-dimensional device, this follows:

$$S_{ELE}(z) = \frac{\phi_{out}[G(z') + \Delta G \cdot \delta(z' - z)] - \phi_{out}[G(z')]}{\int_z \Delta G \cdot \delta(z' - z)} \quad (1.8)$$

Here ϕ_{out} is the PL photon flux induced by the optical generation profile, and ΔG is the additional optical generation induced at depth z' . Using the same principle, the output current density J_{out} and the SCE of a device can be simulated by calculating:

$$S_{CE}(z) = \frac{1}{q} \frac{J_{out}[G(z') + \Delta G \cdot \delta(z' - z)] - J_{out}[G(z')]}{\int_z \Delta G \cdot \delta(z' - z)} \quad (1.9)$$

These simulations allow studying the SELE of devices with different bulk and surface properties and correlating between various recombination mechanisms and the SELE profile features. Simulated profiles of SELE can also be used to validate the extracted SELE profiles and extract quantitative measures from SELE profiles.

Understanding the spatial contribution to external luminescence is crucial for optimizing the performance of optoelectronic devices. The SELE profile quantifies the local efficiency of light emission, reflecting the balance between radiative recombination and non-radiative losses occurring throughout the material. The SELE is suggested as a powerful characterization tool for quantifying the relationships between material properties such as carrier diffusion length and surface recombination velocity and their impact on device performance. This method can be essential for identifying areas where different recombination processes limit overall device performance and to used to map the spatial contribution to the obtainable photo-voltage for different device materials.

1.7 Research Objective

The primary objective of this research is to develop a characterization method that will enable better understanding of the loss mechanisms in solar energy conversion devices. The characterization method is based on the extraction and analysis of the SELE profile of different materials and devices that are common in solar cells or semiconductor devices.

Using both PL measurements and optical modeling, the SELE extraction method is introduced and validated by using finite element simulations. The SELE profile is suggested as a tool for quantifying surface recombination, charge transport across interfaces, and the depth-resolved spatial contribution to the device's photo-voltage. The SELE analysis is further implemented on various materials and samples to demonstrate the analysis of different mechanisms in practical devices. Furthermore, since SELE extraction requires only optical measurements, this method can essentially be used under operando conditions, allowing a direct probe of carrier dynamics and recombination without damaging the sample.

1.8 Thesis Outline

The work presented in this thesis is organized into 4 chapters, each addressing a different aspect of the research. A brief overview of each chapter is provided below.

Chapter 2: Methodology Research and measurement methods are presented in this chapter. It describes the techniques used both experimentally and mathematically, including numerical simulations and experiment setups.

Chapter 3: Results and Discussion In this chapter, the main findings of the research are presented. This chapter starts by presenting results from measurements of GaAs and InP wafers that demonstrate the SELE method. The next sections present the results for SELE analysis conducted on practical PV cells and on MOS capacitors that allow for the investigation of the effects that surface properties have on the SELE. The last part shows the results of an SELE analysis conducted during the formation of a native GaAs oxide layer.

Chapter 4: Conclusion and Future Work The final chapter summarizes the key findings of the research along with recommendations for the implementation of the SELE method in practical applications and its possible impact on PV technologies.

Appendices Appendices A to H provide detailed additional experimental or procedural information that supports this work.

2 Methodology

Under steady-state illumination, optical absorption and charge carrier generation within a device depend on the structure, material properties, and illumination wavelength. Different illumination wavelengths correspond to different absorption depths, therefore, resulting in different charge carrier generation profiles $G(\lambda_{in}, z)$ at each position within the device. At each position, charge transport occurs alongside various recombination processes, including the radiative recombination that generates photons. These photons propagate within the device and can either be reabsorbed or emitted from the device to its ambient environment, thereby contributing to its PL. Typically, high-energy photons (i.e. short wavelengths) are absorbed near the surface, generating carriers that are affected more by surface recombination, while low-energy photons (i.e. long wavelengths) are absorbed deeper within the material, leading to carriers that are influenced primarily by bulk properties or interfaces within the device. Therefore, PL measurements conducted across a broad range of incident wavelengths allow probing the effect of different positions within the device on the radiative recombination processes within the device, ultimately affecting the PL.

This chapter provides a detailed overview of the measurement setups, numerical methods, and simulation models that are used throughout this research. This includes optical modeling techniques, numerical regularization methods, PL simulation, photon recycling model, and details on materials and sample preparation.

2.1 PL and ELE measurements

To measure the photons emitted from a device, two PL measurement setups were constructed. The PL measurement setups were designed to measure the absolute PL signals, with narrow-bandwidth illumination in wide range of wavelengths to achieve high resolution. All measurement instruments and setups described below were purchased, designed, and constructed for the purpose of this study. More details on the design and construction process can be found in G.

2.1.1 PL measurements setup

The measurement setup includes a 75 W Xe lamp light source (Horiba PowerArc). The lamp is coupled to a tunable 320 mm focal length monochromator (Horiba Ihr320) with two possible outputs, together with a 1200 gr/mm with blaze of 500 nm. The system's calculated dispersion is 2.5 nm/mm and achieves a bandwidth of 5 nm with high output illumination power [19]. Generally speaking, the wider the range of illumination wavelengths, the more spatial information can be obtained about the sample, and the smaller the bandwidth of illumination, the better the resolution for the effect of different positions within the device.

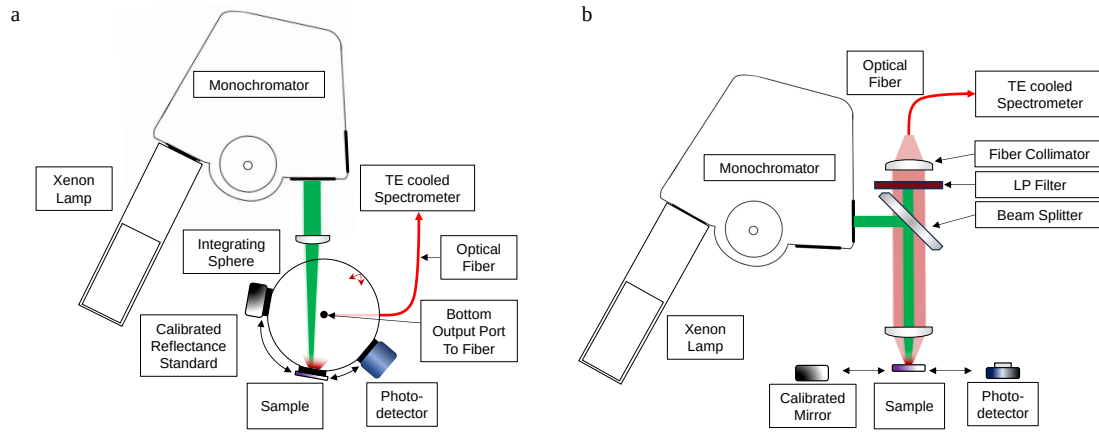


Figure 2.1: (a) PL measurement setup using an integrating sphere. (b) Free space PL measurement setup using a beam splitter and a lens system to collect the PL signal

In each setup, the output monochromatic beam is focused through a set of lenses and mirrors onto the sample. The collection of the PL signal is done with the use of an integrating sphere (IS) or directly through collecting lenses that are located near the sample. In the IS measurement setup, the monochromator output beam is focused on a $4 \times 4 \text{ mm}^2$ spot in the sample port of an integrating sphere (Labsphere RTC-060-SL) which is used to collect the PL signal. The output signal from the integrating sphere is passed through a long-pass filter. The long-pass filter was chosen to correspond to the PL range in order to exclude incident and reflected light from the PL spectrum. The PL signal is then transmitted through an optical fiber (Thorlabs FG910LEC) to a spectrometer (Avantes Avaspec-ULS2048X64TEC-EVO). This measurement setup allows to measure the absolute PL photon flux, as it enables collection from every emission angle. However, using

an integrating sphere highly attenuates the PL signal, as it introduces multiple reflections within the sphere. In the free space (FS) measurement setup, the output beam from the monochromator is directed through a beam splitter and focused onto a $2 \times 2 \text{ mm}^2$ spot on the sample using a 1" lens with a focal length of 35 mm, located about 30 mm from the sample. The PL signal is then directed back through the lens and the beam splitter and is collected using a fiber collimator (Thorlabs CVH100-COL) to the same type of optical fiber and the spectrometer. In this setup, PL signals approximately 100 times weaker were recorded. However, this setup requires a certain calibration to take into account the highly dispersive PL emission. This calibration is detailed in section 2.1.3.

According to (1.6), the incident wavelength-dependent ELE of a sample is found by integrating the PL signal over the emitted wavelengths:

$$\eta_{ext}(\lambda_{in}) = \frac{\int \phi_{out}(\lambda_{PL}, \lambda_{in}) d\lambda_{PL}}{\phi_0(\lambda_{in}) \cdot (1 - R(\lambda_{in}) - T(\lambda_{in}))} \quad (2.1)$$

Where $\phi_{out}(\lambda_{PL}, \lambda_{in})$, is the measured PL spectrum in response to the monochromatic illumination at wavelength λ_{in} , and $R(\lambda_{in})$ and $T(\lambda_{in})$ are the sample reflection and transmission respectively. The incident photon flux, $\phi_0(\lambda_{in})$ was measured prior to the PL measurement, using a calibrated detector (Newport 818-UV/DB), which is placed on the sample's location and plane of incident.

Spectroscopic PL measurements are obtained in counts/nm units. The absolute ELE calculation requires calibrating PL measurements to the total photon flux emitted from the sample. The following section describes the different setup calibration procedures that allow obtaining a quantitative measure of the emitted PL photon flux using the measurement setups. The calibration is performed using NIST traceable calibrated reflectance standards (Labsphere) or a calibrated mirror (Avantes).

2.1.2 Integrating Sphere Setup Calibration

The incident wavelength-dependent ELE measurement setup is shown in 2.1a. The system transfer function, $T_f(\lambda)$, is the ratio between the PL spectrum, $\Phi_{out}(\lambda)$ and the spectrum recorded by the spectrometer, $S_{spec}(\lambda)$. Assuming that the walls of the integrating sphere are perfectly diffusive and that PL photons are emitted isotropically from the sample into the sphere, the transfer function describing PL photons emitted from the sample is identical to that of photons reflected off a diffusive reflectance standard located at the sample port.

Thus, the transfer function is as follows.

$$T_f(\lambda) = \frac{S_{spec}(\lambda)}{\Phi_{std}(\lambda) \cdot t_{int}} \quad (2.2)$$

where $\Phi_{std}(\lambda)$ is the spectrum of photons reflected off the reflectance standard, and t_{int} is the integration time of the spectrometer. The spectrum reflected off the diffusive reflectance standard is:

$$\Phi_{std}(\lambda) = \Phi_0(\lambda) \cdot R_{std}(\lambda) \quad (2.3)$$

Where $R_{std}(\lambda)$ is the reflectance standard reflectivity, and $\Phi_0(\lambda)$ is the incident photon spectrum. When illuminated by the monochromatic light source at wavelength λ_{in} , the input photon flux $\phi_0(\lambda) = \int \Phi_0(\lambda_{in}, \lambda) d\lambda$ can be measured with a calibrated power meter placed at the integrating sphere sample port. Therefore, the transfer function of the system at every wavelength can be found by illuminating a diffusive reflectance standard located at the sample port with a monochromatic beam, recording the resulting spectrum and inserting equation (2.3) into equation (2.2). Since we are interested in obtaining the ELE, this process should be performed at λ_{PL} , which is the set of wavelengths where PL from the sample is expected:

$$T_f(\lambda_{PL}) = \frac{\int_{\lambda_{PL}-\delta\lambda}^{\lambda_{PL}+\delta\lambda} S_{spec}(\lambda) d\lambda}{\phi_0(\lambda_{PL}) \cdot R_{std}(\lambda_{PL}) \cdot t_{int}} \quad (2.4)$$

The integral in equation (2.4) is needed because the input illumination is not perfectly monochromatic, and all the photons reflected off the reflectance standard should be considered. We have taken $\delta\lambda$ to be 15 nm to account for the full spectral width of the input beam, corresponding to the spectrometer resolution of $\pm 10\text{nm}$. For highest accuracy, the reflectance standard should have a reflectance that is similar to the reflectance of the sample, and the monochromator output slit should be as narrow as possible to allow differentiating between different wavelengths.

Once the system transfer function is calculated, the photon emission spectrum in response to illumination at wavelength λ_{in} , can be found:

$$\Phi_{out}(\lambda_{in}, \lambda_{PL}) = \frac{S_{spec}(\lambda_{in}, \lambda_{PL})}{T_f(\lambda_{PL}) t_{int}} \quad (2.5)$$

Integrating $\Phi_{out}(\lambda_{in}, \lambda_{PL})$ over all emission wavelengths results in the total emission photon flux $\phi_{out}(\lambda_{in})$. The incident wavelength-dependent ELE can now be found by inserting equation (2.5) into equation (2.1).

To avoid substitution errors, the sample, reflectance standard, and power meter should be placed on different ports of the integrating sphere when each of them is measured. [20] This ensures that the total reflectance of the sphere is identical in all three experiments (incident power, reflectance standard, and sample measurements).

2.1.3 Free Space Setup Calibration

The calibration of the free space setup was performed against the calibration of the integrating sphere configuration. To calibrate the system, the PL measurements for every incident wavelength were performed for a GaAs wafer sample in both the calibrated integrating sphere setup and the free space setup. Then both measurements were used to calculate a calibration function between the two setups. In general, the free space setup allowed recording signals that were up to 100 times smaller compared to the integrating sphere setup.

2.1.4 Emission Angle Calculation

To detect the PL signal in the free space setup, the angle of emission of the PL should be taken into account. Especially in samples with specular surfaces, photon propagation can undergo total internal reflection (TIR) and reduce the total emitted photon flux. This phenomenon generates an emission cone in which photons are emitted, resulting in non-isotropic emission from the sample. TIR occurs when light propagates from a high-refractive-index material to a lower-refractive-index medium (e.g., air). For simple cases, the critical angle for total internal reflection is defined by Snell's law and equals:

$$\theta_c = \sin^{-1}(n_1/n_2) \quad (2.6)$$

where n_2 is the refractive index of the material and n_1 is that of the ambient medium. When radiative recombination-generated photons propagate to the surface at angles greater than θ_c , they are reflected internally rather than transmitted through the interface. Only photons that are emitted in angles lower than the critical angle can thus be emitted from the material. The range of angles for which photons can be emitted is termed the escape cone. The escape cone restricts the emitted light to a narrow angular range centered around the surface normal, and the majority of the generated light undergoes multiple internal reflections and may be reabsorbed or trapped, depending on the material geometry and optical design. The escape cone and angle θ_c of the layered interfaces were calculated using the transfer matrix method (TMM), which is described in Section 2.2.3.

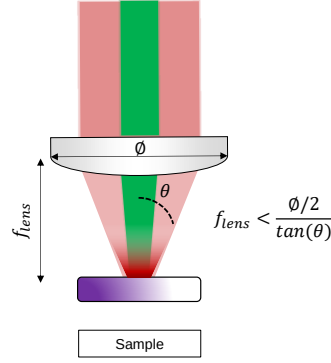


Figure 2.2: Free space PL measurement setup details. The green color represents incident light wavelength and the red color represents the PL emission from the sample. The PL emission occur in a cone, corresponds to the TIR angle of the sample. The lens was chosen with large aperture and strong focal length to accept a large angle of PL emission. The setup was calibrated against the integrating sphere setup where isotropic PL signal is collected. Details on the escape cone calculations can be found in B

Thus, the free space measurement setup was designed to obtain PL signals that are emitted with a large angle. The focusing lens, which also serves as the objective lens, was therefore chosen to accept a large angle of PL emission, on account of the focused illumination beam size. The lens was chosen to have a large aperture ($\phi = 25.4mm$) together with a strong focal length of $f_{lens} = 35mm$. This corresponds to a collection angle that is larger than 20° , as can be seen in figure 2.2. For a specular sample, the escape cone for PL emission was calculated with the half-angle of $\approx 17^\circ$, thus the setup is designed to collect all emitted photon. B. This design of the measurement setup allowed to measure high signal-to-noise ratio PL signals. However, this does not eliminate the necessary calibration against the integrating sphere setup, where all the PL signal is collected.

2.2 Optical Characterization and Modeling

2.2.1 Absorption Measurements

The absorption of the sample was determined using spectrophotometric reflection and transmission measurements that were performed with UV-Vis spectrophotometers (Agilent Cary-5000, UMA and PerkinElmer LAMBDA 1050 +, TAMS) with an incident angle of the sample of 8° , which is the standard incident angle for measurements and is similar to the incident angle of the sample on the sphere. Using reflection and transmission, together with incident flux measurements, the absorbed photon flux $\phi_{abs}(\lambda_{in})$ can be obtained, as described in 2.1:

$$\phi_{abs}(\lambda_{in}) = \phi_0(\lambda_{in}) \cdot (1 - R(\lambda_{in}) - T(\lambda_{in})) \quad (2.7)$$

2.2.2 Optical Constants Measurements

The optical constants of a sample were obtained by using spectroscopic ellipsometry measurements (J.A. Woollam M-2000 DI) and analyzed with Variable Angle Spectroscopic Ellipsometer (VASE) software. The SE measurements were conducted at incident angles of 65° , 70° , and 75° degrees of incident, and the ellipsometer reflection intensity was calibrated against a known Si/SiO₂ standard. The GaAs and InP wafer samples were modeled as a stack of two layers, consisting of a GaAs or InP substrate and a native oxide surface layer.[21] Tabulated values for the optical constants were used to find the oxide layer thickness while comparing the results to the reflection measurements. The optical constants were then fitted to find the correct deviation from the tabulated values. The GaAs and InP optical constants were extracted while accounting for free carrier absorption (FCA) using the Drude model in the software. The model assumed a homogeneous and specular sample with no absorption in the interface between the oxide layer and the ambient. The extracted optical constants were verified against both reflectance and transmittance measurements of the sample together with a comparison of the spectral emission using PL simulations. More details on the verification method can be found in Appendix B

Ellipsometry Model

The ellipsometry model consisted of the GaAs substrate and a native oxide surface layer. An iterative fitting procedure was followed to find the best fit between the spectroscopic

ellipsometry experimental data, the native oxide thickness, and a GenOsc model used for the substrate. The GenOsc model enables the construction of a material dielectric function by assembling different oscillator parameters [22]. The GenOsc model consists of one Drude oscillator, six PSemi oscillators, and one Gaussian oscillator. The extracted optical constants are presented in Figure B.1. The Drude oscillator parameters used in this model are shown in Table 2.1.

Table 2.1: Drude oscillator parameters used in the ellipsometry analysis

Parameter	Parameter Input
Doping [cm^{-3}]	$1 \cdot 10^{19}$
Mobility [cm^2/Vs] ($\mu_{majority}$)	101.68 [23]
Effective mass (m^*)	0.1 [23]

2.2.3 Transfer Matrix Method (TMM)

The optical transfer matrix method is used to calculate the reflection and transmission properties at optical interfaces between a sample and its environment. In this method, each interface in the structure of a device is characterized by a matrix that relates incident and reflected electric fields, as well as phase accumulation. When light encounters multiple and thin interfaces, net reflection and transmission coefficients are determined by multiplying the individual matrices in sequence. Assuming unpolarized light, reflection, absorption and optical generation calculations are performed using the Burkhard et al. TMM Matlab code, while inserting the measured optical constants for substrates and oxides and the tabulated optical constants for different layers. [24] For angle dependent fresnel coefficients, that are used in PL simulation, K. J. Pescoe and S. Divitt Matlab codes were used as a base for calculation. [25, 26].

2.2.4 Carrier Generation profile

Using the TMM method, the measured optical constants from the ellipsometry analysis are used to calculate the absorption coefficient α and the carrier generation profile for each incident wavelength $G(\lambda_{in}, z)$.

The optical constants for different GaAs samples with various dopants and doping concentrations were tested. Within the incident wavelength range, ($\approx 400 - 800[nm]$) the

optical constants did not show substantial differences between samples with different doping concentrations and dopants. A significant difference in optical constants, specifically in the extinction coefficient, κ , occurs around the band-gap energy of the material. The extinction coefficient is crucial for the calculation of the spontaneous radiative recombination rate and the PL simulations. Therefore, special attention is needed for the correct extraction and verification of the optical constants in the wavelength range that corresponds to the materials band-gap.

2.3 MOS Capacitor Characterization

To characterize the properties of a MOS capacitor, capacitance voltage (C-V) measurements were conducted using an LCR meter. In this measurement, the sample is subjected to a voltage sweep ranging from negative to positive values, while a small AC signal is applied to measure the capacitance at each bias.

The capacitance of MOS capacitor varies with the bias voltage applied to the metal gate. When the voltage is swept, the MOS capacitor undergoes changes in capacitance, reflecting different operational regions: accumulation, depletion, and inversion. Typical C-V curves exhibit distinct characteristics depending on the doping type (p-type or n-type) and the measurement frequency, offering insight into both the doping and the oxide layer thickness and permittivity.

Hysteresis measurement can also be used to characterize MOS capacitors. Hysteresis measurements assess the response of the sample to cycling bias voltages and is used to identify capacitance properties due to charge trapping. The bias voltage is varied quickly in both directions, and the capacitance is recorded. The presence of a hysteresis loop in the C-V plot can indicate charge trapping and characterize the performance of the dielectric layer.

Impedance measurements are used when capacitance measurements are unsuccessful due to large or small series or parallel resistances. The LCR meter C-V measurements usually obtain the impedance by fitting the results to an equivalent circuit model and deriving the assumed capacitance. When both series and parallel resistances are substantial, the capacitance can be found by analyzing the impedance measured over a wide range of frequencies.

In this general case, the impedance Z of the circuit can be described as:

$$Z(\omega) = R_s + \frac{R_p}{1 + j2\pi\omega CR_p} \quad (2.8)$$

where ω is the small signal angular frequency, R_s and R_p are the series and parallel resistances, and C is the capacitance of the MOS capacitor. Measurement of impedance over a wide range of frequencies allows fit to the capacitance and resistances of the circuit using this equation.

2.4 SELE Extraction methods

SELE profiles are mathematically derived from ELE measurements of a device or sample under wide illumination wavelengths using equations (1.5) and (1.7). In this process, equation (1.7) is first discretized:

$$\overline{\eta}_{ext}(\lambda_{in}) = \frac{1}{\overline{\phi}_{abs}(\lambda_{in})} \overline{\overline{G}}(\lambda_{in}, z_i) \overline{S}_{ELE}(z_i) \quad (2.9)$$

In this case, the device thickness was divided into one-dimensional spatial elements of various widths such that $\overline{S}_{ELE}(z_i)$ is a vector of the average SELE within the element z_i , and $\overline{\overline{G}}(\lambda_{in}, z_i)$ is a matrix that describes the total optical generation within the element located at position z_i that is induced by monochromatic illumination with wavelength λ_{in} . The matrix $\overline{\overline{G}}(\lambda_{in}, z_i)$ is calculated as the total optical generation rate within an element with boundaries at z_i and z_{i+1} , as follows:

$$\overline{\overline{G}}(\lambda_{in}, z_i) = \frac{z_{i+1} - z_i}{2} (G(\lambda_{in}, z_i) + G(\lambda_{in}, z_{i+1})) \quad (2.10)$$

2.4.1 Standard Tikhonov Regularization

To extract the SELE, a Tikhonov regularization process is used, similar to what was previously shown for the extraction of SCE.[7, 6] Since equation (2.9) has an infinite number of solutions, the candidate solutions must be constrained. The measured incident wavelength-dependent ELE, and the modeled optical generation profile, $\overline{\overline{G}}(\lambda_{in}, z_i)$, are inserted into equation (2.9), and the extracted SELE vector, $\overline{S}(z)$, is found to minimize the following expression:

$$\min \left\{ \left\| \frac{G(\lambda_{in}, z) \overline{S}_{ELE}(z)}{\overline{\phi}_{abs}(\lambda_{in})} - \overline{\eta}_{ext}(\lambda_{in}) \right\|^2 + \kappa^2 \|L(\overline{S}_{ELE}(z))\|^2 \right\} \quad (2.11)$$

Where $\|\cdot\|$ is the euclidean norm (sometimes written as $\|\cdot\|_2$). The first term in the brackets defines the accuracy in which the candidate solution, $\overline{S}_{ELE}(z)$, reproduces the measured ELE as described in equation (2.9). L is an operator such that the second term in the brackets constrains the physical properties of the solution.

Small values of κ produce solutions that accurately reconstruct the measurement, but may result in a non-physical solution. Large values of κ result in over-constrained solutions. These solutions might have a physically acceptable shape and magnitude, but cannot

reconstruct the measured data with an acceptable accuracy. The desired solution is a compromise between solutions that are constrained to have physical features and solutions that reconstruct the measurements accurately.

Different operators for L effect the results and the output of the solution. For example, when L is the identity operator, the constraint is on the magnitude of the solution. This reduces the magnitude of the solution at points where such a reduction does not lead to a large deviation from the measured values. For instance, in regions where the optical generation is very low (i.e. insufficient information is provided). When L is chosen as the first or second derivative operators, the solutions are constrained to be smoother. The regularization parameter κ is a weight used to ensure that the solution reproduces the measured ELE accurately while maintaining physical features according to the chosen L operator. The candidate solutions are SELE vectors that minimize (2.11), each corresponding to a different value of κ . From this group of solutions, only those that produce both a small deviation from measurements and correspond to the physical constraint are averaged to produce the mean profile, which is the extracted SELE.

For every value of κ , the reconstruction error $\varepsilon = \left\| \frac{\bar{G}(\lambda_{in}, z_i) \bar{S}_{ELE}(z)}{\phi_{abs}(\lambda_{in})} - \bar{\eta}_{ext}(\lambda_{in}) \right\|^2$, and the constrained candidate solution, $\|L(\bar{S}_{ELE}(z))\|^2$ are calculated. The L-curve is a plot of ε as a function of $\|L(\bar{S}_{ELE}(z))\|^2$. The knee of the curve corresponds to values of κ that lead to solutions that both have physical characteristics and can reproduce the measurements with an acceptable accuracy. κ_{knee} is the weight variable that corresponds to the knee of the L-curve. The chosen profile is the mean of the solutions with κ values in the range of $\sqrt{10}\kappa_{knee} < \kappa < \kappa_{knee}/\sqrt{10}$.

When the sample thickness is considerably larger than the absorption depth for all incident wavelengths, optical generation is negligible in significant portions deep within the device. The optical generation matrix $\bar{G}(\lambda_{in}, z_i)$ in these regions is practically zero, leading to high uncertainty in the reconstructed $\bar{S}_{ELE}(z)$ deep within the sample. Thus, the regularization process cannot provide accurate SELE results for regions where an insignificant charge carrier occurs. This issue is insignificant for practical solar cells where the optical generation is non-zero at every depth within the device. For this reason, PL measurements from the back surface can provide additional information for the SELE on the other side of the sample. When the PL signal from the back surface is substantial, the SELE should therefore not be zero at the edge of the sample. This can be done by performing PL measurements in "transmission" mode by illuminating one surface of the sample and measuring PL off the other.

The choice of mesh affects the extracted SELE profiles and must be designed with great attention. Near the surface, where most of the charge generation occurs, the mesh is uniform. Deeper within the wafer, where the optical generation is significantly lower, the size of the mesh elements follows a logarithmically spaced array. The logarithmically spaced mesh is chosen so that its first element is in the size of the uniform linear mesh elements. The length of the linear mesh is chosen according to the maximum absorption depth which corresponds to the highest incident wavelength. The number of elements in the linear mesh is chosen to produce the lowest RMSE between the ELE measurement and the reconstruction according to equation (1.7). The number of elements in the second mesh was chosen as the smallest number that results in a non-negative SELE throughout the material.

2.4.2 Wide-Range Tikhonov Regularization

The standard Tikhonov regularization method described above may produce sets of solutions that correspond to non-physical SELE profiles. The SELE is defined as a probability parameter; therefore, only solutions between values of 0 and 1 are meaningful. For this reason, a modified regularization process can be used. In this method, the regularization is performed for a large set of weights, κ , and considers solutions that are only between 0 and 1 at every point. This set of solutions is then averaged to provide the mean SELE solution. Here, each solution corresponds to a different value of κ . Thus, due to the large set of solutions, this modified regularization process is termed here as wide-range Tikhonov regularization.

2.4.3 Projected Tikhonov Regularization

Another modification can be performed to account for the expected results, where the solution for $S_{ELE}(z)$ must be within 0 and 1 for all values of z . In this approach, each element of the standard Tikhonov regularization solution $S_{ELE}(z)$ is forced to 0 when negative and to 1 when larger than 1. The projection operator is defined as:

$$P(S(z)) = \max(0, \min(1, S(z))) \quad (2.12)$$

This method essentially forces the solution to exist only in the $[0, 1]$ interval. This method is a simple constraint that falls under a large group of projection techniques. For this reason, this modified regularization process is termed the Projected Tikhonov regularization.

2.4.4 Projected Gradient Descent Regularization

The projected gradient descent (PGD) method can also be used to solve the minimization problem that is directly derived from equation (2.9): [27]

$$\min \left\{ \left\| \overline{\overline{G}}(\lambda_{in}, z_i) \overline{S}_{ELE}(z_i) - \overline{\phi_{abs}}, (\lambda_{in}) \overline{\eta_{ext}}(\lambda_{in}) \right\|^2 \right\} \quad (2.13)$$

In this case, the values of $S(z)$ are iteratively updated in the direction of the negative gradient, so that in each iteration i , the updated value for the solution follows:

$$S^{(i+1)} = S^{(i)} - \alpha \nabla f(S^{(i)}) \quad (2.14)$$

Where $S^{(i)}$ represents the value of the solution $S(z)$ in iteration i , α is a coefficient that represents the progression rate for each iteration (that is, the learning rate) and $\nabla f(S^{(i)})$ is the gradient of the problem, with respect to $S(z)$ in iteration i , which means:

$$\nabla f(S^{(i)}) = 2 \overline{\overline{G}}(\lambda_{in}, z_i)^T \cdot (\overline{\overline{G}}(\lambda_{in}, z_i) S^i - \overline{\phi_{abs}}, (\lambda_{in}) \overline{\eta_{ext}}(\lambda_{in})) \quad (2.15)$$

To account only for the results where the solution for $S(z)$ must be within 0 and 1 for all values of z , each step incorporates a projection step to enforce this constraint. The projection operator is defined in the previous section. The projected gradient descent is applied to solve equation (2.9) where $S(z)$ is the SELE vector that is constrained between 0 and 1.

The loss function ε is calculated for each iteration, and the learning rate is reduced by a small factor when the loss function stays the same after two or more iterations. The change in learning rate allows for finding accurate solutions over many iterations.

To account for the measurement error in the input ELE vector, which can affect the resulting solution, the PGD method is applied multiple times and a random error of $\pm 10\%$ is introduced for the ELE measurements for every iteration. The final results were averaged to obtain the mean SELE solution.

2.4.5 Inverse Laplace Regularization

For simple devices or structures, optical generation and absorption can generally be described by the Beer-Lambert absorption law. Using this assumption, previous studies have demonstrated the extraction of SCE by applying the inverse Laplace transformation to

equation (1.1) [28]. Similarly, the same analysis can be performed to extract the SELE. In this case, the optical generation is assumed to follow the Bear-Lambert law. Thus, simplified optical generation can be described as follows:

$$G(\lambda, z) = \alpha(\lambda)\phi_0(\lambda)A(\lambda)e^{-\alpha(\lambda) \cdot z} \quad (2.16)$$

Where ϕ_0 is the incident photon flux, A is the absorption of the sample, and α is the absorption coefficient. Equation (1.6) can therefore be described with respect to α :

$$\eta_{ext}(\alpha) = \frac{1}{\phi_{abs}(\alpha)} \int_0^d \alpha\phi_0(\alpha)A(\alpha)e^{-\alpha \cdot z}S(z)dz \quad (2.17)$$

Assuming that the carrier generation originates only from incident photons, the absorbed photon flux ϕ_{abs} is therefore equal to the incident flux multiplied by the absorption ϕ_0A . Thus, the relation can be simplified and written as follows:

$$\frac{\eta_{ext}(\alpha)}{\alpha} = \int_0^d e^{-\alpha \cdot z}S(z)dz \quad (2.18)$$

In the case of $d \rightarrow \infty$, the normalized external luminescence efficiency becomes the Laplace transform of the SELE, $S(z)$.

$$\mathcal{L}\{S\}(\alpha) = \frac{\eta_{ext}(\alpha)}{\alpha} \quad (2.19)$$

Therefore, under these assumptions, the SELE can be extracted using the inverse Laplace transformation.

Here, the measured absorption coefficient α , as well as ELE measurements with respect to α are taken. Then, the quantity for the Laplace transform of S is calculated following (2.19). In order to perform the inverse Laplace transformation, the calculated quantity is fitted to a continuous function which is a sum of two exponents. This fitting is performed for all values of α in the incident wavelength range. The inverse Laplace transformation is then performed on the fitted continuous function and is evaluated in significant values of z . To perform this inverse Laplace transformation numerically, both the Euler and Talbot, de-Hoog, and Gave-Stehfest methods were tested [29, 30, 31, 32, 33] .

2.5 SELE Simulation

2.5.1 Finite Elements simulations

SELE profiles were computed in simulations by using equation 1.8. To simulate the PL emission spectrum while accounting for charge carrier drift, diffusion, generation and recombination processes, the semiconductor continuity equations for electrons and holes were coupled to the Poisson equation and to an optical model while considering the redistribution of photons through photon recycling.

The coupled Poisson and continuity equations for electrons and holes were solved numerically using a commercial finite-elements partial differential equations solver (COMSOL Multiphysics V6.1). The simulation domain for all models is one-dimensional with a uniform cross section of 1cm^2 . The boundary conditions applied were a reference potential of 0 V on the back surface and an applied voltage $V_{\text{applied}}[\text{V}]$ on the front surface. In the case of homogeneous wafers under no external voltage, a zero-field boundary condition was applied. Trap-assisted surface recombination was also used as the boundary condition for electrons and holes on both surfaces. The material properties and the optical generation profile produced by the incident monochromatic illumination, $G(z)$, are inputs to the simulation. To account for degeneracy effects, Fermi-Dirac statistics are assumed. Radiative, Auger, and SRH recombination were assumed for the bulk material.

For relatively thin samples and models, the mesh element lengths were chosen to be uniform. However, with larger samples, such as a $350\mu\text{m}$ thick wafer, the length of the mesh elements of the model was chosen to follow a logarithmic scale due to the increased concentration of carriers near the surface. The doping concentration and thickness of the model are based on the sample properties as provided by the manufacturer. Deviations from measured and expected values might be due to inaccuracies in the optical absorption coefficients obtained from the ellipsometry measurements. Because of the sensitivity of the PL to the absorption coefficient at energies near the band gap, any deviation in the absorption coefficients results in a relatively large deviation in other material properties.

The SELE of the device was then calculated using equation (1.8), where the additional optical generation at point z , $\Delta G \cdot \delta(z - z')$ was added to the optical generation profile induced by the background illumination. To assure that the delta-function generation is only a small signal and does not have a significant effect on the energetics of the device,

the size of ΔG is set to be 10% of the total optical generation in the sample:

$$\int_0^d \Delta G \cdot \delta(z - z') dz = 0.1 \cdot \int_0^d G(z) dz \quad (2.20)$$

2.6 PL simulation

Under the assumption of isotropically radiated photons, the electron and hole concentrations were used to calculate the spontaneous emission rate per solid angle at every depth, following: [34]

$$dR_{rad}(z, E) = \frac{1}{4\pi^3 h^3 c_0^2} \alpha_b(E) n(E)^2 E^2 \frac{1}{\exp\left(\frac{E - (E_{F_C}(z) - E_{F_V}(z))}{k_B T}\right) - 1} \quad (2.21)$$

where n is the sample's refractive index, and E_{F_C} and E_{F_V} are the electrons and holes quasi-fermi levels, and $\alpha_b(E)$ is the energy dependent absorption coefficient. Although light absorption is attenuated by FCA in the bulk; the generation of photons in spontaneous emission is determined by wavelength-dependent absorption coefficients that are not affected by FCA.[34] We therefore distinguish between the absorption coefficient of the bulk, that relates to light emission, α_b , and the absorption coefficient, α , that includes FCA.[35]

The spatial distribution of the quasi-Fermi levels as well as the concentration distribution of electrons and holes at every depth is found by solving the continuity equations thus accounting for the effects of recombination and transport mechanisms.

Radiated photons are either reabsorbed, which contributes to the photo-generation or emitted through the front or back surface, thus contributing to the PL signal. The contribution of reabsorbed photons to the optical generation is calculated in an iterative process that produces the spontaneous emission rate, which accounts for photon recycling within the material. The PL spectrum from the front or back surface of the device is then calculated using the corrected spontaneous emission rate multiplied by photon escape probabilities, following:

$$\phi_{out_{sim}}^{front/back}(E) = \int_0^d \int_0^{2\pi} \int_0^\pi dR'_{rad}(z, E) \cdot P_{esc}^{front/back}(z, E, \theta, \phi) \cdot \sin(\theta) d\theta d\phi dz \quad (2.22)$$

Where $P_{esc}^{front/back}$ is the probability for a photon to escape from the front or back surface. E is the photon energy, z is the depth from the surface in which the photon was generated,

θ is the photon propagation angle with respect to the normal of the surface, and ϕ is the azimuthal angle. Once the photon flux is calculated, the incident wavelength-dependent ELE and the SELE can then be calculated using equation (1.6) and equation (1.8), respectively. Escape probabilities are calculated following the method described in Steiner et al. [36]. Surface reflection and transmission calculations were implemented using the TMM method as described in Section 2.2.3.

2.6.1 Rough surfaces consideration

Optical modeling and measurements of rough surfaces require special attention. As rough surfaces can create non-specular reflections and dispersion of light from the surface, this can affect ellipsometry measurements and PL emission modeling. For this reason, double-sided polished samples were used and the reflection from each surface was assumed to be specular in the optical models. Nonetheless, various tools (such as the GenPro4 software) are available for calculating optical generation, absorption and reflection, in samples with certain surface roughness. [37, 38]

2.7 Photon recycling model

When charge carriers recombine radiatively, the photons that are generated within the material propagate until they are reabsorbed or until they reach an interface of which they are reflected or transmitted. Photons that are transmitted to the ambient contribute to the PL spectrum and the ELE. Therefore, to accurately simulate the emitted PL flux, photon re-absorption (photon recycling) and its affect on the optical generation should be accounted for. The effects of photon recycling were simulated by coupling a ray optics simulation to a finite element simulation, which models charge transport throughout the device (Section S5.). The coupling of the two simulations was performed by using an iterative algorithm similar to the method previously reported by Walker et al. [39]. The optical model used

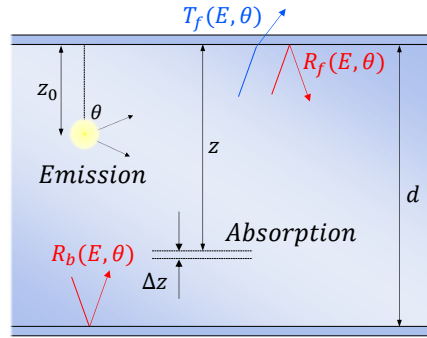


Figure 2.3: Photon emission and absorption schematics in a 2D infinitely wide device of thickness d . A photon is generated at a distance z_0 from the front surface and absorbed in a small depth Δz at a distance z . $R_f(E, \theta)$, $R_b(E, \theta)$ and $T_f(E, \theta)$ are the front and back reflectance and transmission Fresnel coefficients, respectively.

for the photon recycling simulation assumes a flat and homogeneous material of thickness d , with a thin native oxide layer on front and back surfaces, with no absorption in the oxide layers. Figure 2.3 shows a schematic diagram of the processes that photons undergo within the material. The model assumes Beer-Lambert optics, and that the generated photons propagate isotropically. Thus, the fraction of photons that are absorbed in a small element with thickness Δz , that is located at a distance z from the front surface, prior to

any reflections, are:

$$A^{dir}(\theta, z, z_0, E) = \frac{\alpha_b}{\alpha} \begin{cases} e^{-\frac{\alpha(E)(z_0-(z+\Delta z))}{\cos(\theta)}} - e^{-\frac{\alpha(E)(z_0-z)}{\cos(\theta)}} & \cos(\theta) > 0 \\ e^{-\frac{\alpha(E)(z-z_0)}{|\cos(\theta)|}} - e^{-\frac{\alpha(E)((z+\Delta z)-z_0)}{|\cos(\theta)|}} & \cos(\theta) < 0 \end{cases} \quad (2.23)$$

Where θ is the angle between the photon propagation direction and the normal to the front surface, and z_0 is the depth in which the photon was generated. The pre-factor α_b/α is used to account for photon absorption events that contribute to the generation of mobile charge carriers and exclude free carrier absorption (see section S2.). The fraction of photons that are reflected once prior to being absorbed in Δz is:

$$A_1^{ref}(\theta, z, z_0, E) = \frac{\alpha_b}{\alpha} \begin{cases} \left(e^{-\frac{\alpha(E)(z_0+z)}{\cos(\theta)}} - e^{-\frac{\alpha(E)(z_0+(z+\Delta z))}{\cos(\theta)}} \right) R_f(\theta, E) & \cos(\theta) > 0 \\ \left(e^{-\frac{\alpha(E)(2d-z_0-(z+\Delta z))}{|\cos(\theta)|}} - e^{-\frac{\alpha(E)(2d-z_0-z)}{|\cos(\theta)|}} \right) R_b(\theta, E) & \cos(\theta) < 0 \end{cases} \quad (2.24)$$

$R_f(\theta, E)$ and $R_b(\theta, E)$ are the angle dependent reflectivity of the front and back surfaces respectively. $R_f(\theta, E)$ and $R_b(\theta, E)$ were calculated using the TMM method. The absorbance of photons undergoing two reflection events is:

$$A_2^{ref}(\theta, z, z_0, E) = \frac{\alpha_b}{\alpha} \begin{cases} \left(e^{-\frac{\alpha(E)(2d+z_0-(z+\Delta z))}{\cos(\theta)}} - e^{-\frac{\alpha(E)(2d+z_0-z)}{\cos(\theta)}} \right) R_f(\theta, E) R_b(\theta, E) & \cos(\theta) > 0 \\ \left(e^{-\frac{\alpha(E)(2d-z_0+z)}{|\cos(\theta)|}} - e^{-\frac{\alpha(E)(2d-z_0+(z+\Delta z))}{|\cos(\theta)|}} \right) R_b(\theta, E) R_f(\theta, E) & \cos(\theta) < 0 \end{cases} \quad (2.25)$$

This analysis is repeated for higher orders of reflection, to account for the contribution of photons with energies that are near the band gap energy and are hardly absorbed. At $\theta = \pm 90^\circ$, $A^{dir} = \alpha_b/\alpha$ for $z = z_0$, and $A^{dir} = 0$ for all other values of z . Since at $\theta = \pm 90^\circ$ photons are not reflected off any surface, $A_i^{ref}(\pm 90^\circ, z_0, z, E) = 0$. Thus, the probability for a photon generated at z_0 to be reabsorbed in an element Δz at depth z from the front surface, $P_{PR}(z, z_0, E)$, is the absorptance at z accounting for all reflection orders and propagation angles.

$$P_{PR}(z, z_0, E) = \int_0^{2\pi} \int_0^\pi A^{dir} + \sum_i A_i^{ref} d\theta d\phi \quad (2.26)$$

The normalization by 2π is required to account for the isotropic emission of radiative recombination photons. The optical generation due to photons that are reabsorbed at depth z , $G_{PR}^i(z)$, then follows:

$$G_{PR}^i(z) = \int_0^\infty \int_0^d \frac{1}{\Delta z} dR_{rad}^i(E, z_0) P_{PR}(z, z_0, E) dz_0 dE \quad (2.27)$$

Where $dR_{rad}^i(E, z)$ is the photon generation rate calculated according to equation (2.21). The superscript i is the photon recycling order as explained below. The total optical generation rate is the generation rate induced by the incident illumination, $G_0(z)$, with the contribution from photon recycling:

$$G^i(z) = G_0(z) + G_{PR}^i(z) \quad (2.28)$$

The optical generation due to photon recycling augments the electron-hole distribution, which in turn determines the distribution of photons generated through radiative recombination. Thus, the continuity and Poisson equations must be coupled to the equations describing photon recycling. This coupling was implemented with an iterative process as illustrated in Figure 2.4. The Poisson and continuity equations are solved using a COMSOL multiphysics simulation. From this simulation, the separation of the depth-dependent quasi-Fermi levels is calculated and inserted into equation (2.21), which is used to compute an updated value for $dR_{rad}^i(E, z_0)$. Equation (2.27) is then used, together with $P_{PR}(z, z_0, E)$, to calculate an updated value of the optical generation contribution, $G_{PR}^i(z)$. The updated optical generation profile $G^i(z)$ is inserted back into the COMSOL simulation and the charge carrier distribution is recalculated. This process is repeated until convergence. Once the solutions converge, the PL spectrum and the ELE can be calculated according to

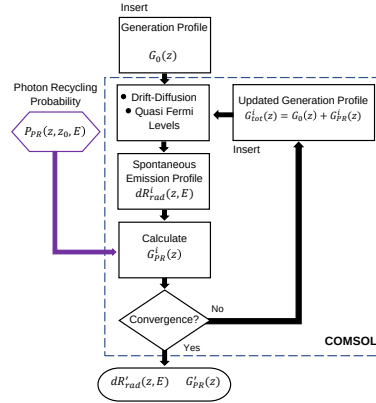


Figure 2.4: A flow chart of the iterative process for coupling photon recycling with the charge transport simulations.

equation (2.22). The total photon escape and reabsorption probabilities follow:

$$P_{abs}(z, E) = \int_0^d \frac{1}{\Delta z} P_{PR}(z, z_0, E) dz_0 \quad (2.29)$$

$$P_{esc}(z, E) = 1 - \int_0^d \frac{1}{\Delta z} P_{PR}(z, z_0, E) dz_0 \quad (2.30)$$

Since PL measurements are conducted in reflection or transmission mode, $P_{esc}(z, E)$ should be divided to the escape probability from the front and back surface ($P_{esc}^{front}(z, E)$ and $P_{esc}^{back}(z, E)$ respectively) following Steiner et al. [36]

$$P_{esc}^{front}(z, E) = \int_0^{2\pi} \int_0^\pi \frac{T_f(\theta, E) \left(e^{-\frac{\alpha(E)z}{\cos(\theta)}} + R_b(\theta, E) e^{-\frac{\alpha(E)(2d-z)}{\cos(\theta)}} \right)}{1 - R_f(\theta, E) R_b(\theta, E) e^{-\frac{\alpha(E)2d}{\cos(\theta)}}} \sin(\theta) d\theta d\phi \quad (2.31)$$

$$P_{esc}^{back}(z, E) = \int_0^{2\pi} \int_0^\pi \frac{T_b(\theta, E) \left(e^{-\frac{\alpha(E)(d-z)}{\cos(\theta)}} + R_f(\theta, E) e^{-\frac{\alpha(E)(d+z)}{\cos(\theta)}} \right)}{1 - R_b(\theta, E) R_f(\theta, E) e^{-\frac{\alpha(E)2d}{\cos(\theta)}}} \sin(\theta) d\theta d\phi \quad (2.32)$$

$T_f(\theta, E)$ and $T_b(\theta, E)$ are respectively the front and back surface angle-dependent transmittance for a photon with energy E , calculated the same way as $R_f(\theta, E)$ and $R_b(\theta, E)$. More details on escape probabilities and the photon recycling model are shown in the AppendixC.

2.8 Materials and Sample Preparation

To demonstrate the SELE extraction method, a model sample was chosen for measurements. The samples for the initial development of the methods were chosen to be simple homogeneous wafers, as they were easier to optically model and their properties are less affected by other factors such as internal electric field or losses at interfaces.

In the initial phase of this study, two main materials were tested for this purpose. InP was first tested as a possible material for a model sample. InP photoelectrodes are commonly used in various photoelectrochemical processes, where surface states and treatments greatly affect their performance. Therefore, InP is a good candidate for SELE analysis, as this could provide insight into the surface properties of the anodes and the material. However, InP samples showed an unstable PL signal over time, which yielded inconsistent results. More details on the results for InP can be found in Appendix A. Consequently, this led us to continue with a GaAs wafer as a model sample for the initial development of the SELE extraction method.

After successful results with the GaAs wafer, we continued with the SELE analysis of GaAs-based devices. These allowed studying the SELE of complex device structures. These devices include MOS capacitors and layered samples, with different doping levels and thickness.

2.8.1 GaAs wafers

GaAs is a key material in opto-electronics and advanced solar cells. It is a direct band gap material with high carrier lifetimes, which can therefore produce high- and stable-PL signals. For these reasons, it is most commonly used in optical sensors, LEDs, multi-junction as well as in concentrated PV and for space applications. In such devices, the surface and interface properties directly affect cell efficiency and performance. For this reason, GaAs was chosen as a material for the model sample for the development of the SELE methodology. Doping can affect the PL signal of the sample. High doping increases possible recombination centers and can increase the PL intensity.

2.8.2 Double sided polish wafers

In this work, all experiments were performed on double-sided polished wafers. Using a double-sided polished wafer is important in ellipsometry measurements and PL emission modeling. The two-sided polish wafers help avoid unwanted scattering or depolarization from a rough backside that can affect ellipsometer measurements, where the analysis assumes smooth interfaces and well-defined optical paths. A rough or unpolished backside can introduce errors or artifacts in the measured signal, especially for transparent or semi-transparent materials, where light can reflect from both surfaces. Such errors are critical especially with high doping samples, affecting the

results for higher wavelengths and the optical constants fit that is used in PL calculations and simulation. Details on optical constants measurements and validation for highly doped samples are discussed in Appendix B.

2.8.3 Native Oxides of GaAs wafers

GaAs oxidize in air to form a thin layer of oxides on the surface of the exposed material. The interactions of Ga and As atoms with oxygen in the air form different oxides with mainly Ga_2O_3 , As_2O_3 compounds on the surface, with different stoichiometry. [40] These oxides can then interact with the substrate material to form a thin layer of As near the surface. [41] This oxide layer generally acts as a weak passivation layer by filling up dangling bonds at the surface of the sample. Therefore, PL measurements can be used to identify the presence of a native oxide layer on the surface of GaAs.

The thickness of the native oxide layer usually follows a logarithmic rate law that eventually stabilizes to form a thin layer of 2 to 5 nm on the substrate surface. [2] The thickness of the oxide layer was modeled using the relation following Lukes et al. [2]

$$d = a_0 + b_0 + \ln(t) \quad (2.33)$$

Here, d is the thickness of the layer in nm , the pregrowth factors are $a_0 = 0.5[nm]$ and $b_0 = 0.62[nm/decade]$. The thickness of the layer was then used in comparison with the SELE results as well as to accurately model the reflectance of the sample when applying the TMM

For purposes of device fabrication, pretreatment of the substrate surface is key for creating good electrical contacts or isolation layers. Thus, the native oxide layer is usually removed pre-processing. [42] Chemical etching is commonly used to remove the native oxide layer. Different chemicals can be used to remove native GaAs oxides with acidic compounds such as HF and HCl , or alkaline compounds such as $(NH_4)OH$. [43] One notable study has even suggested specific attention to etching procedures in the context of GaAs solar cell fabrication and final device performance. [44]

The formation of oxide layers depends on many variables such as the wafer crystalline orientation, temperature, moisture, DI water resistivity, and external irradiation. [2, 45, 46] PL measurements were performed over time using short illumination times of 0.1 seconds for each illumination wavelength, which is small compared to the measurement time. The affect of illumination and illumination wavelength on the oxide layer formation rate can thus be neglected. [47]

Sample Preparation

The sample investigated in this section was a p-type GaAs wafer (100) that underwent chemical etching prior to PL measurements to study the effect of native oxide formation on PL and SELE. The sample was first cleaned to remove any organic residues from the surface of the substrate. The samples were placed in acetone (3 min), then in DI water (1 min), followed by isopropanol (3 min), DI water (1 min), and dried using N_2 . The sample was then etched using a 1:1 mixture of HCl and DI water (37% by weight mixed 1:1 in H_2O) for 60 seconds, to remove the native surface oxides and contaminants. After HCl etching, the sample was dipped in DI water and dried using N_2 . The sample was immediately transferred to the PL measurement setup for measurements. Note that PL measurements made immediately after surface treatment suggested that *HCl* performed better removal of native oxide than $(NH_4)OH$. The samples that were treated with *HCl* showed a significant change in PL with time, suggesting the formation of native oxides and evidence of their removal. This is compared to samples that were treated with $(NH_4)OH$, following a protocol by Yang et al. [48] that did not show any change in PL after etching.

2.8.4 MOS Capacitors

GaAs MOS capacitors were fabricated in the available facilities at the university "center for nano-science and technology". The MOS capacitor was designed to enable PL measurements under various applied voltages. For this reason, Indium Tin Oxide (ITO), which is a transparent and conductive material, was selected for the top gate. These properties enable illumination and transmission of the emitted PL signal combined with electrical contact.

Sample Preparation

The p-type GaAs wafer (100) was also used as the substrate for the MOS capacitor. The sample was first cleaned to remove any organic residues from the surface of the substrate. The samples were placed in acetone (3 min), then in DI water (1 min), followed by isopropanol (3 min), DI water (1 min), and dried using N_2 . The MOS capacitor substrate etched using a 1:1 mixture of HCl and DI water (37% by weight mixed 1:1 in H_2O) for 60 seconds, to remove the native surface oxides and contaminants. After HCl etching, the sample was dipped in DI water and dried using N_2 . The sample was transferred to the MOS capacitor oxide layer deposition or the contact layer deposition within 5 minutes of native oxides removal.

No post-annealing was performed on the oxide layer, as some post-annealing (performed in $600^\circ C$ for 30 s under N_2 and also under O_2) has introduced some bubble-like features on the surface and destroyed the uniformity of the oxide layer.

To create the dielectric layer of the MOS capacitor, an isolating oxide layer of Alumina (Al_2O_3) of 20 nm was then deposited on the wafer surface using ALD deposition (BENEQ TFS200). On top of the oxide layer, to provide a transparent and conductive layer, a thin 100 nm Indium Tin Oxide layer (ITO) was deposited using PVD sputtering (VINCI TECHNOLOGIES PVD 20). The sample was then subjected to rapid thermal annealing (RTA) at 450°C for 5 min, to anneal the ITO layer and improve its electrical properties. And finally, the back contact was deposited by sputtering Ti, Pt, and Au layers with thicknesses of 10, 5 and 100 nm, respectively. Performing the RTA process after the bottom contact did not show significant differences in the contacts' performance. Samples of approximately 5 mm by 5 mm were cut from the wafer to reduce the probability of shunts due to pinholes in the oxide layer and improve the sample properties.

A diagram of the final MOS capacitor structure can be seen in figure 2.5.

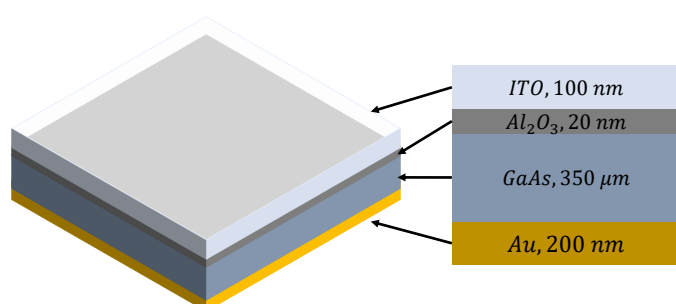


Figure 2.5: Diagram depicting GaAs MOS capacitor structure

A standard operating procedure (SOP) document was prepared and shared to ensure that the fabrication steps were followed as intended. The SOP document is included in the Appendix F.

3 Results and Discussion

3.1 Results overview

This chapter presents the experimental and simulation results obtained from the investigation of the SELE of various semiconductor materials and devices. The results first describe the SELE of a GaAs wafer sample on which the method was developed and established. The SELE results of an InP wafer are then shown and compared to the SELE results of the GaAs wafer. SELE simulations of P-N junctions are then shown, presenting the potential application of SELE analysis on practical devices. Next, the SELE of MOS capacitors under different bias voltages is presented. This results shows the effect of surface properties on the SELE profile. The final section details the dynamic SELE during the formation of GaAs's native oxide layer. The experimental results and SELE analysis are arranged from basic wafers to complex devices and then dynamic processes to demonstrate the development of the method.

3.2 SELE of a GaAs Wafer

The SELE extraction method was developed using a model sample. For this, a simple and homogeneous GaAs wafer was chosen. In this section, measurements were performed on a double-sided polished, $(1.74 \pm 0.83) \cdot 10^{19} \text{cm}^{-3}$ Zinc-doped, $350 \mu\text{m}$ thick GaAs wafer (Wafer Technology Ltd.).

The SELE of GaAs wafers was extracted in three steps: first, an optical model was constructed to calculate the wavelength-dependent optical generation profile $\Delta G(\lambda_{in}, z)$, then the wavelength-dependent ELE was obtained from PL measurements; and finally, the SELE was extracted according to equation 1.7. The optical constants of the samples were obtained with spec-ellipsometry analysis, and the optical generation profile was calculated using the TMM, as described in the methods section. The incident wavelength-dependent PL spectra were measured in the integrating sphere setup. Figure 3.1 shows the incident wavelength-dependent PL spectra measured. Figure 3.2a shows the incident wavelength-dependent ELE spectrum calculated using the measured PL spectra and equation 1.6. The SELE profile of the GaAs wafer was extracted from this ELE spectrum and the optical model using Tikhonov regularization as described in the methods section.

Figure 3.2b shows the extracted SELE profile near the surface, and the inset shows the profile deeper within the wafer. The solid curve represents the mean profile that was obtained from all solutions that are both physically constrained and can reconstruct the measurements. The shaded area is the standard deviation of all these solutions.

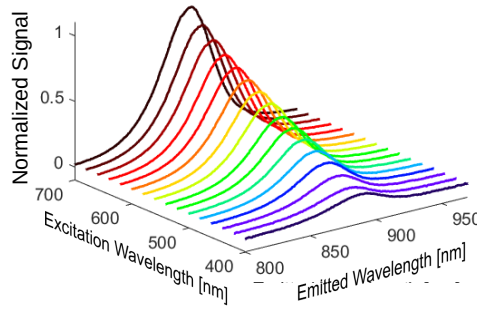


Figure 3.1: PL measurements of p-type GaAs sample for different excitation wavelengths.

The SELE profile maps the competition between radiative and non-radiative recombination processes, as well as the probability for photons that are generated within the bulk of the device to be emitted. Near the surface, the low SELE is a result of surface recombination through defects suppressing radiative recombination processes. As a result, the SELE initially increases with the distance from the surface. However, deeper within the material, radiative recombination photons are less likely to be emitted due to self-absorption, and the SELE decreases. A considerable portion of the photoluminescence photons are at energies slightly below the band gap. Since these photons are hardly reabsorbed even when generated far from the surface, the decay of the SELE with the distance from the surface is significantly more moderate than dictated by the absorption coefficient for photons with energy higher than the band gap. The non-zero SELE deep within the sample was validated experimentally with transmission mode PL measurements. In these experiments, the input beam is incident on one surface of the sample, but only photons emitted off the opposite surface are collected by the integrating sphere. The inset in Figure 3.2a shows the PL spectra measured in reflection mode and transmission mode for an incident wavelength of 600nm. Both spectra were normalized by the maximum value of the reflection mode PL spectrum. The non-zero PL measured in transmission mode implies that optical generation near one surface leads to photon emission off the other. Thus, the SELE is higher than zero even near the back surface of the sample and consequently all through the device.

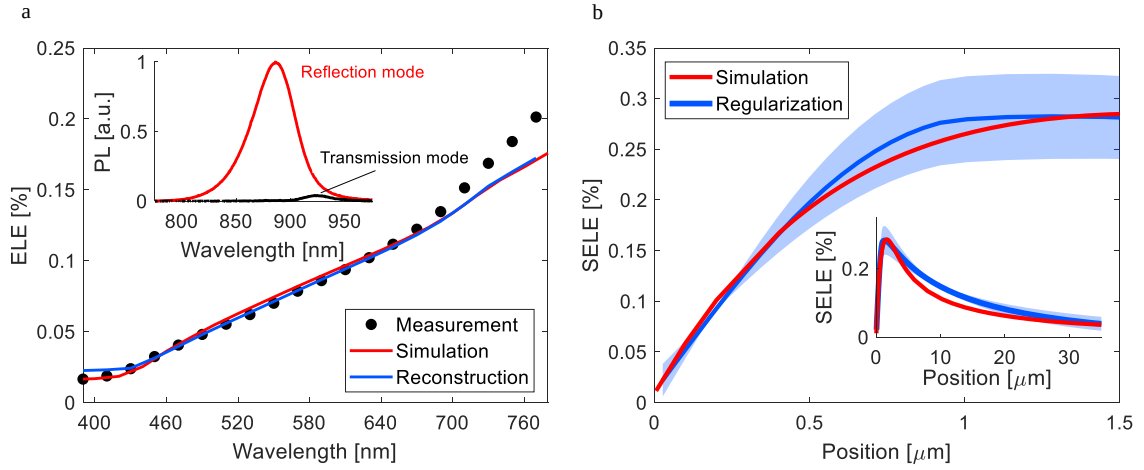


Figure 3.2: ELE spectrum and SELE profile of a p-type GaAs wafer. (a) The incident wavelength-dependent ELE measurement, simulation, and reconstruction from the extracted SELE profile. ELE measurements were taken in reflection mode. The inset shows a comparison of reflection and transmission mode PL measurements for an incident wavelength of 600 nm. The simulation parameters were set to fit best the ELE measurement. (b) Simulated and extracted SELE profile. The shaded area is the standard deviation of all the candidate solutions.

3.2.1 SELE Simulation

The extracted SELE profile was validated by reconstructing the incident wavelength-dependent ELE spectrum and by finite element simulation. The reconstructed ELE spectrum is calculated by inserting the extracted SELE and optical generation profiles into equation (1.7). The resulting ELE spectrum is shown in Figure 3.2a. Optical modeling errors and inadequate regularization procedures can result in deviations between the measured and reconstructed spectra. [6, 7] The excellent agreement between the spectra supports the validity of the regularization process and the extracted SELE profile.

The extracted SELE profile was further investigated using device simulations in which the emitted PL spectrum can be calculated for any optical generation profile and material properties. The incident wavelength-dependent ELE spectrum was calculated by providing the simulation with optical generation profiles corresponding to monochromatic illumination inputs, and the SELE is found using delta function-like optical generation profiles according to equation (1.8). The properties of the sample material were fitted to best match the emitted PL spectrum and the incident wavelength-dependent ELE. The simulated ELE is shown in Figure 3.2a and a comparison between the simulated and measured PL spectra can be found in the Appendix B. The material properties used in this simulation are listed in Table 3.1. Figure 3.2b shows the simulated SELE profile. The agreement between the simulated and extracted profiles further validates the SELE extraction method.

3.2.2 Effect of Material Properties on SELE Profile

To correlate between different features in the SELE profile and material properties, SELE profiles were simulated for GaAs wafers with optical properties similar to those of the measured sample, but with various surface recombination velocities and Shockley Reed Hall (SRH) lifetimes (all the other material properties are as in Table 3.1). Figure 3.3a shows the simulated SELE profiles with different surface recombination velocities. The bulk lifetime for minority carriers is $0.6ns$. As expected, a higher surface recombination reduces the SELE at the surface, while the SELE within the device is unaffected by the change in the surface recombination velocity. In the extreme case of a long diffusion length and very high surface recombination velocity, the transport of charge carriers that are generated as deep as $5\mu m$ from the surface is still affected by surface recombination. The inset shows the SELE at the surface as a function of the surface recombination velocity. The SELE at the surface is extremely sensitive to the surface recombination velocity for surface recombination velocities above $10^4 cm/s$. Thus, in this range, the effect of surface treatments can be quantified by analyzing the SELE at the surface. For surface recombination velocities below $10^4 cm/s$, surface recombination is not the dominant recombination process, and the SELE at the surface is determined by other mechanisms. Figure 3.3b shows the computed SELE profiles for various minority carrier SRH lifetimes and a surface recombination velocity of $4 \cdot 10^7 cm/s$. Increasing the lifetime of the minority carrier increases the SELE in the bulk of the device because of the increase in the internal luminescence efficiency. However, due to the high surface recombination velocity, the SELE at the surface remains almost unchanged. The drop in the SELE towards the surface and the position of the SELE peak are a result of surface recombination of charge carriers that are generated within the bulk of the material and diffuse towards the surface where they recombine. Thus, when surface recombination is the dominant loss mechanism, the position of the maximum value of the SELE is comparable to that of the diffusion length. The inset in Figure 3.3b shows the effective diffusion length, $L_{eff} = \sqrt{D\tau_{eff}}$, and the maximum position of the SELE as a function of the SRH lifetime. Here, it is assumed that the effective lifetime follows $\tau_{eff}^{-1} = \tau_{SRH}^{-1} + \tau_{Rad}^{-1} + \tau_{Aug}^{-1}$ where τ_{Rad} is the radiative lifetime and τ_{Aug} is the Auger lifetime. Both lengths show a very similar trend that saturates for high values of τ_{SRH} where the radiative lifetime becomes the dominant factor determining the effective lifetime of the minority carrier and the diffusion length.

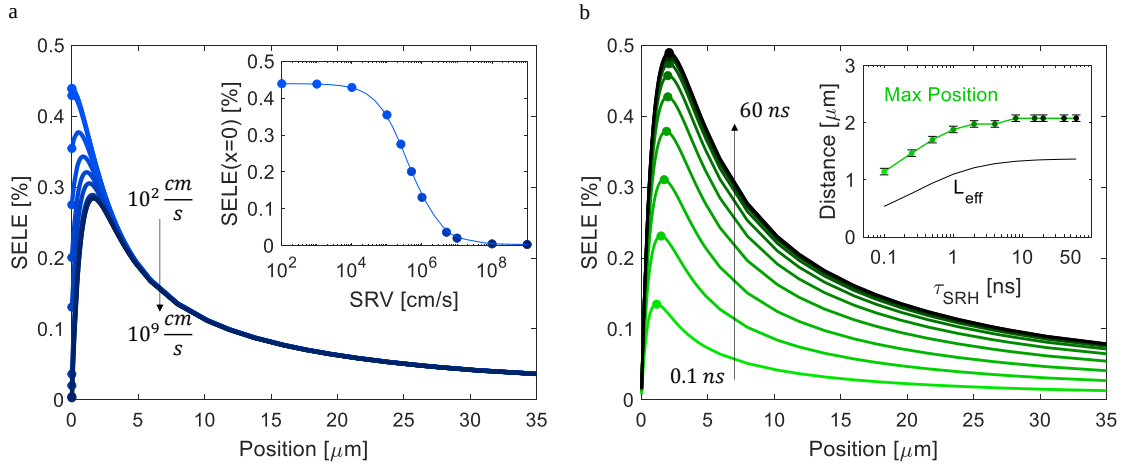


Figure 3.3: Simulated SELE profiles of p-type GaAs wafers. (a) SELE profiles for wafers with surface recombination velocity between 10^2 cm/s and 10^8 cm/s and τ_{SRH} of 0.6 ns . The inset shows the SELE at the surface as a function of the surface recombination velocity. The color-coding of the markers matches that of the SELE profiles. (b) Simulated SELE profiles for wafers with τ_{SRH} between 0.05 ns and 40 ns and a surface recombination velocity of $4 \cdot 10^7 \text{ cm/s}$. The inset shows the peak position of each curve and the calculated effective diffusion length. The error bars correspond to the width of simulated element at the maximum position. All the other input parameters used for the simulation are detailed in section S5 (Supporting Information).

Material Properties in Simulation

Table 3.1. lists all the material properties used in the simulation. The material properties were initially taken from the referenced values, and those were modified to find the best fit between the ELE measurements and the simulation.

Parameter	Best fit	Referenced Value {Samples spec.}
Doping [cm^{-3}]	$1 \cdot 10^{19}$	$\{(1.74 \pm 0.83) \cdot 10^{19}\}$
Thickness [μm]	350	$\{364 \pm 2\}$
Mobility [cm^2/Vs] (majority, minority)	(101.68[23], 1500)	$\{71 \pm 10\}, 1000$ [23]
Radiative recombination coefficient [cm^6/s]	$2 \cdot 10^{-10}$	$1 \cdot 10^{-10}$ [49]
Auger recombination coefficient [cm^6/s]	$15 \cdot 10^{-30}$	$10 \cdot 10^{-30}$ [50]
SRH lifetime [ns] (holes, electrons)	(0.4, 0.4)	(0.82, 1.9) [51]
Surface Recombination velocity [cm/s]	$4 \cdot 10^7$	$1 \cdot 10^7$ [52, 53, 54]
Band gap narrowing model	Jain – Roulston[55]	-

Table 3.1: GaAs material properties used in simulation

3.2.3 Photo-voltage Contribution of the SELE

The buildup of the obtainable photo-voltage within a device is determined by the contribution of different regions to the external luminescence. According to equation (1.4), the deviation from an ideal open circuit voltage follows $\Delta V_{oc} = V_{oc} - V_{oc,ideal} = \frac{kT}{q} \ln(\eta_{ext})$. Inserting equation (1.7) into the expression for ΔV_{oc} yields:

$$\Delta V_{oc}(d) = \frac{kT}{q} \ln \left(\int_0^d \frac{\Delta G(z)S(z)}{\phi_{abs}} dz \right) \quad (3.1)$$

The term in the integral can be described as the spatial contribution to the photo-voltage, and the buildup of the photo-voltage within the device can be found by integrating this term over various device thicknesses. The spatial contribution to the photo-voltage, that can be obtained with an absorber layer that is made of the tested GaAs, was estimated using the extracted SELE profile from Figure 3.2b and an optical generation profile corresponding to AM1.5 illumination [56]. The obtainable photo-voltage buildup, $\Delta V_{oc}(d)$, was then calculated using equation (11) for various absorber layer thicknesses. Figure 3.4 shows the spatial contribution to the photo-voltage (left vertical axis), and $\Delta V_{oc}(d)$ for different absorber layer thicknesses (right vertical axis). Almost the entire photo-voltage is built within $2\mu m$ next to the surface, providing an upper bound for the thickness of an absorber layer made of this material. Also shown in Figure 3.4 are the photo-voltage contribution and ΔV_{oc} calculated for a material with negligible surface recombination ($S = 100$ cm/s). The difference in ΔV_{oc} between the two curves allows estimating the voltage losses due to surface recombination. For example, for a thickness of $0.5\mu m$, the ΔV_{oc} difference is 54 mV and for a thickness of $2\mu m$, the ΔV_{oc} difference is 41 mV.

Since the photo-voltage buildup with depth is estimated by integrating the same SELE profile over different thicknesses, it does not account for changes in the SELE due to recombination at the assumed back surface. As shown in Figure 3.3a, high surface recombination velocity can reduce SELE as far as $5\mu m$ from the surface. Hence, the $\Delta V_{oc}(d)$ buildup in Figure 3.4 serves as an upper bound for the photo-voltage assuming an ideal back surface.

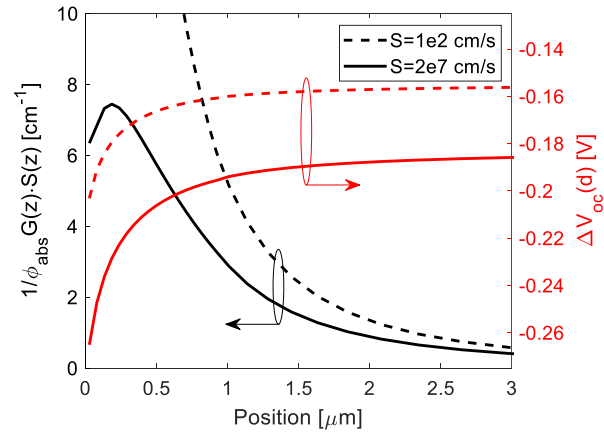


Figure 3.4: The spatial contribution to ΔV_{oc} (left vertical axis), and ΔV_{oc} as a function of the device thickness (right vertical axis) when the surface recombination is significant ($S = 4 \cdot 10^7 \text{ cm/s}$, solid curves) and when the surface recombination is negligible ($S = 100 \text{ cm/s}$, dashed curves). The optical generation profile corresponds to AM 1.5 illumination and the SELE profile is as in Figure 3.2b and Figure 3.3a.

3.3 SELE of an InP Wafer

Initial testing for ELE measurements showed that all InP samples experienced some PL instability over time. However, when the illumination of the sample was followed by a sufficiently long dark period, PL measurements were successfully reproducible. For this reason the PL measurements described in this section were recorded during one-second illumination, followed by a three-minute period in the with no illumination. Details on the PL transient processes that occur during illumination can be seen in appendix A .

Measurements were performed on a single side polished, $(1.4 \pm 0.08) \cdot 10^{17} \text{ cm}^{-3}$ S-doped, $350 \mu\text{m}$ thick InP wafer (Wafer Technology Ltd.). The SELE of the InP wafers was extracted in the same method as was described for the GaAs wafer in the previous section. First, the optical generation profile $\Delta G(\lambda_{in}, z)$ was calculated using the optical constants from ellipsometry and spectroscopic measurements. The ELE profile was then obtained from PL measurements; and finally, the SELE was extracted according to equation (1.7). The incident wavelength-dependent PL spectra were measured in the free space PL system as shown in 2.1b. Figure 3.5a shows the incident

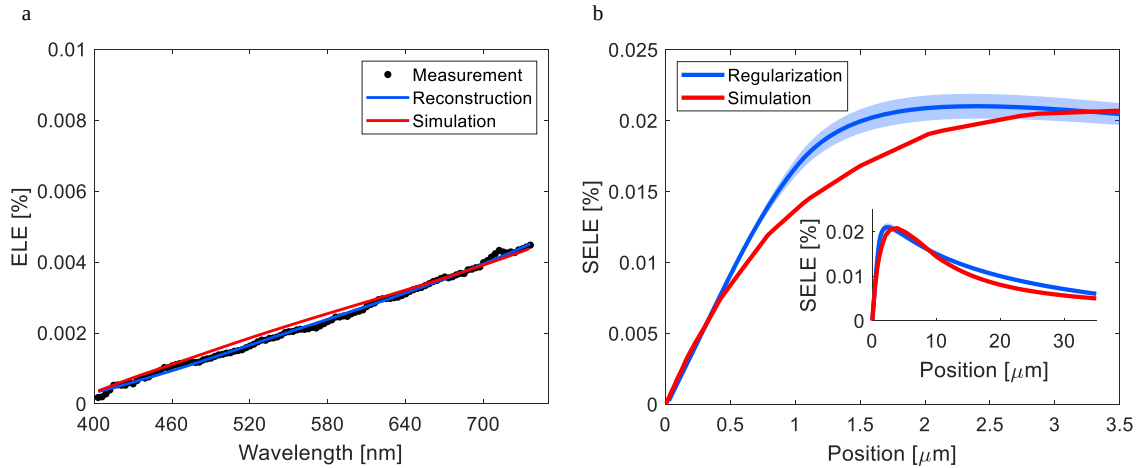


Figure 3.5: ELE and SELE profiles of a n-type InP wafer . (a) The incident wavelength-dependent ELE measurement, simulation, and the spectrum reconstructed from the extracted SELE profile. ELE measurements were taken in reflection mode. (b) Simulated and extracted SELE profile. The shaded area is the standard deviation of all the candidate solutions.

wavelength-dependent ELE spectrum calculated using the measured PL spectra and equation (1.7). The SELE profile of the InP wafer was extracted from this ELE spectrum and the optical model using Tikhonov regularization, in the exact same method that was performed for GaAs, as described in the methods section. Additionally, finite element simulation was performed using a simulation method similar to that described for GaAs. Figure 3.5b shows the extracted SELE profile near the surface, and the inset shows the profile deeper within the wafer. The solid curve represents the

mean profile that is obtained from all solutions that are physically constrained and can reconstruct the measurements. The shaded area is the standard deviation of all these solutions.

3.3.1 SELE of InP and GaAs

Compared with the SELE results for GaAs, the SELE peak of InP is found deeper within the sample at $4.5\mu\text{m}$, compared to $1.5\mu\text{m}$ for GaAs. Additionally, the slope of the SELE profile near the surface is larger than that for GaAs. These features indicate a higher carrier mobility and carrier lifetime in InP, as discussed in figure 3.3. The large mobility and lifetime allow photo-generated carriers to diffuse further from the surface and away from non-radiative surface recombination centers, resulting in a larger overall diffusion length for InP. This difference in lifetimes and mobilities can also be seen in the energy band diagram generated from the simulation, shown in figure 3.7. In this figure, the longer diffusion length and mobility in InP result in a higher difference in the quasi-Fermi levels and a higher concentration deeper within the sample.

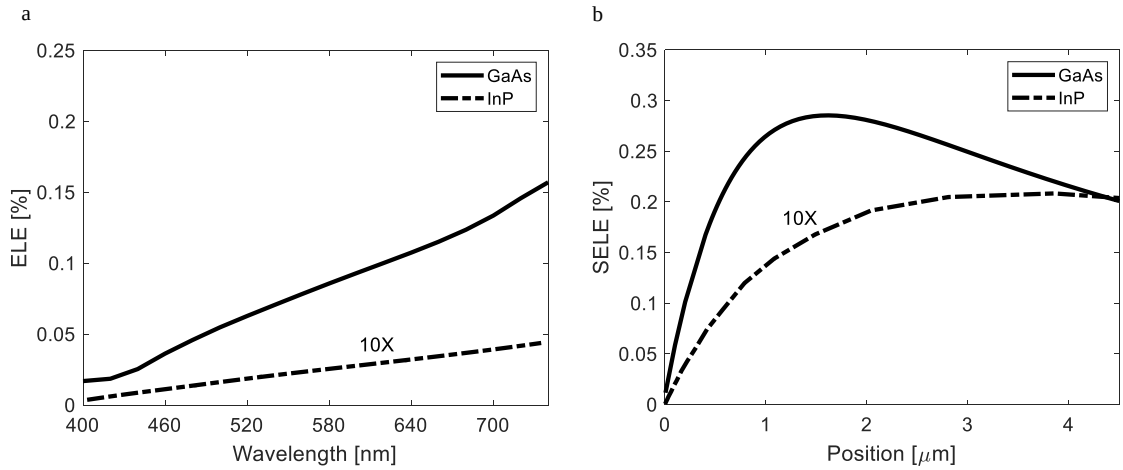


Figure 3.6: ELE and SELE simulation results of GaAs and InP wafers, the results can be seen individually in 3.2 and 3.5 respectively. The ELE and SELE results for InP are shown multiplied in a factor of 10 for comparison

These properties can explain the small InP ELE compared to that of GaAs. The long lifetime results in a larger carrier concentration deeper within the material. The carrier concentration, or the difference in quasi-Fermi levels, is directly related to the radiative recombination rate. Thus, when radiative recombination eventually occurs deeper within the material, the generated photons are more likely to be reabsorbed in the material, leading to lower ELE and SELE.

The high surface recombination velocity obtained from the fitted the simulation is estimated to be smaller in practice and is probably the result of the relatively weak PL signal, especially for the short wavelength illumination range. Incident photons with an energy that corresponds to this

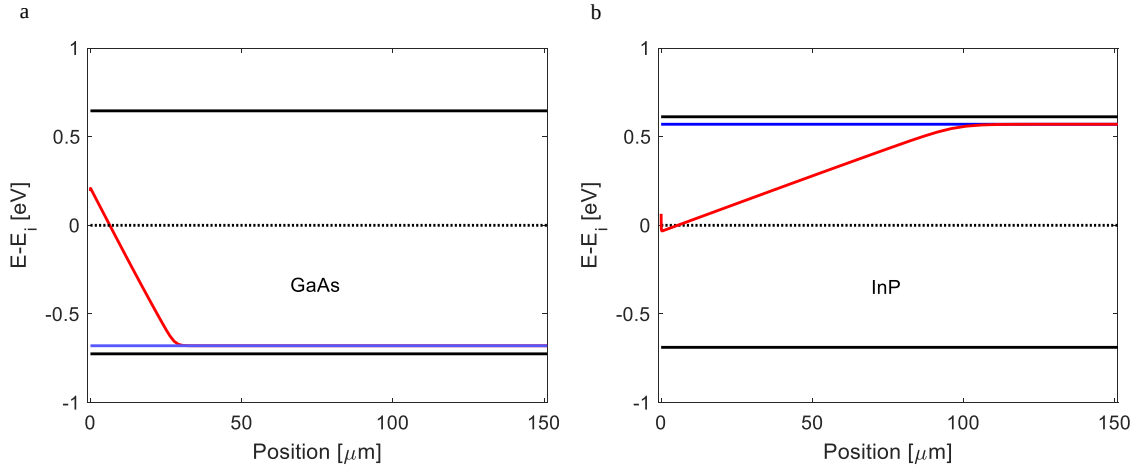


Figure 3.7: Energy band diagrams of GaAs (a) and InP (b) wafers from simulation under the same illumination, the lifetime and mobility properties follow 3.1 and 3.2 respectively

Parameter	Best fit	Referenced Value {Samples spec.}
Doping [cm^{-3}]	$1.0 \cdot 10^{17}$	$\{(1.4 \pm 0.08) \cdot 10^{17}\}$
Thickness [μm]	350	$\{362 \pm 7\}$
Mobility [cm^2/Vs] (majority, minority)	(3500, 140)	$\{3260 \pm 80\}, 120$ [23]
Radiative recombination coefficient [cm^6/s]	$1.5 \cdot 10^{-10}$	$\approx 1.0 \cdot 10^{-10}$ [57]
Auger recombination coefficient [cm^6/s]	$15 \cdot 10^{-31}$	$9 \cdot 10^{-31}$ [58]
SRH lifetime [ns] (holes, electrons)	(150, 300)	300[57, 59]
Surface Recombination velocity [cm/s]	$2 \cdot 10^7$	$< 1 \cdot 10^5$ [60]
Band gap narrowing model	Jain – Roulston[55]	-

Table 3.2: InP material properties used in simulation

range are mainly absorbed within the first few tens of nanometers. Therefore, accurate PL results within that range is critical for correct SELE results near the surface. For InP, the measured PL flux was $\approx 0.06 \mu W/cm^2$, which is five orders of magnitude lower than the illumination intensity, and close to the sensor sensitivity. This resulted in higher noise in the PL signal which affected the ELE results and the SELE near the surface. Additionally, the mentioned transient PL effects that occurred during measurements were especially strong in the lower wavelength illumination range, which affected measurements.

3.3.2 Material Properties in Simulation

Table 3.2. lists all the material properties used in the simulation. The material properties were initially taken from the referenced values, and those were modified to find the best fit between the ELE measurements and the simulation.

3.4 SELE of PN junctions

After the SELE analysis was established using both measurements and simulations on simple wafers, this section extends the method to a practical solar cell device structure. Unlike wafers, solar cells introduce different layers and carrier transport dynamics that affect luminescence behavior. By applying SELE simulation to such device structures, the SELE is demonstrated as a characterization tool for solar cells and thin-film solar cells.

3.4.1 Thin-film GaAs solar cell model

In this section the SELE simulations of a thin-film GaAs solar cell structure is shown. The device structure chosen here is similar to that of a thin film solar cell that was previously described by Walker et al. [39] The device structure is shown in table 3.3.

Additionally, the effect of photon recycling on the I-V curve and ΔV_{oc} is calculated using the photon recycling model described in the method section. These results were compared with the results in the study by Walker et al. where a photon recycling model was used in correspondence with device measurements. The comparison between the present findings and those of Walker et al. serves to validate the photon recycling model.

The optical generation profile within the thin film structure was first calculated using TMM. Then, the optical generation profile was used as an input for the simulation. The optical constants for GaAs were taken from ellipsometry measurements of the GaAs sample used previously, and the optical constants for the other materials were taken from available literature and detailed in table 3.3.

The reflectivity of the cell and the charge carrier generation profile were calculated assuming 1 sun illumination using the standard AM1.5G, normalized to $1000W/m^2$, as in the original study. [56] The cell's reflectivity and carrier generation profile are shown in figure 3.8. In this model, the top window and ARC layers were treated as thin layers with zero absorptance, and the bottom layer is assumed to be perfectly reflective. Therefore, both the finite elements and photon escape probabilities calculations included only the emitter, base and back surface field layers, where the top window and ARC layers are considered as a boundary layers.

First, the I-V curves were calculated and compared with the I-V curve results in the Walker et al. study. [39] The I-V curves were simulated with and without the photon recycling model, as shown in figure 3.9. The summary of results can be seen in table 3.4.

The large difference in the maximum current density between this and Walker's study is due to the optical model and recombination properties. Here, the surface recombination between the window/emitter and the window/ARC layers is not included in the simulation but is considered in Walker's model. Moreover, despite using the same sources for optical constants, Walker's study

Table 3.3: PN junction structure adopted in optical generation calculations. The Layers in bold represents the layers adopted in finite elements simulations. Material properties were similar to the material properties in Walker et. al.

Layer	Material	Thickness [μm]	Optical constants ref.
<i>ARC1</i>	<i>MgF</i>	0.082	[61]
<i>ARC2</i>	<i>TaO</i>	0.053	[62]
<i>Window</i>	<i>n - AlInP</i>	0.025	[63]
<i>Emitter</i>	<i>n - GaAs</i>	0.120	
<i>Base</i>	<i>p - GaAs</i>	3.5	
<i>Back surface field</i>	<i>p - Al_{0.05}Ga_{0.95}As</i>	0.40	[64, 65]
<i>Mirror</i>	<i>Silver</i>	0.100	[66]

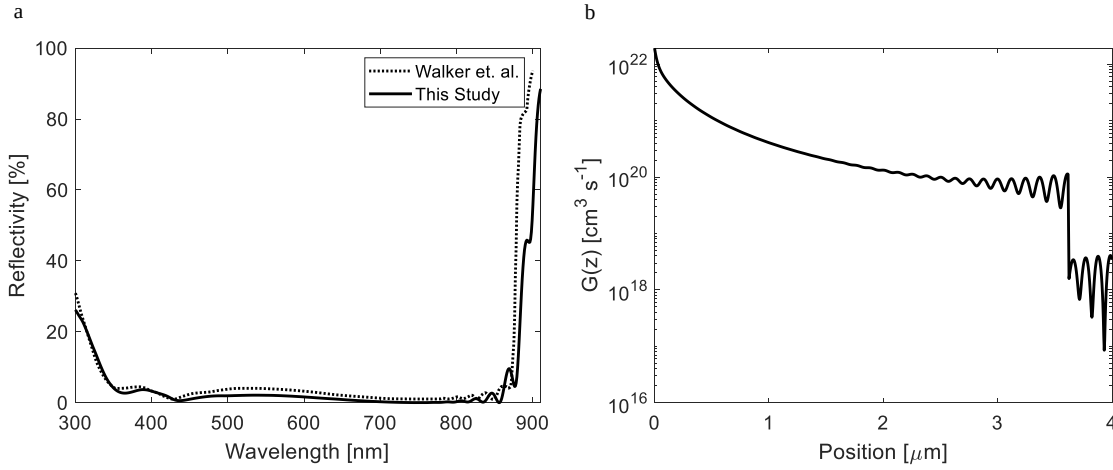


Figure 3.8: (a) Thin-film GaAs solar cell reflectivity calculated using the TMM method compared to measured cell reflectivity in Walker et. al. study. The details of the cell structure can be seen in table 3.3 (b) The calculated carrier generation rate profile for AM1.5 illumination of the simulated cell. The fluctuations are a result of interference within the material

noted that a certain optimization of the dielectric function and optical constants in the ARC layers was performed, in order to match the measured and modeled reflectivity of the cell. These changes can be the source for the difference in reflectivities, as shown in figure 3.8, and can also be the source for the difference in the total current density.

However, the results for open-circuit voltage with and without photon recycling showed similar results. The addition of the photon recycling processes to the model resulted in an increase of $0.034[V]$ to the open circuit voltage using our model, which was similar to the $0.036[V]$ obtained by Walker. The open circuit voltage is defined by the carrier concentration. Since photon recycling contributes to an increase in carrier concentration, it affects mainly the obtainable open circuit voltage. Thus, the similar difference between the open circuit voltage results with and without the

use of the photon recycling model, ΔV_{oc} , compared to the results of the study by Walker can be seen to validate our photon recycling approach.

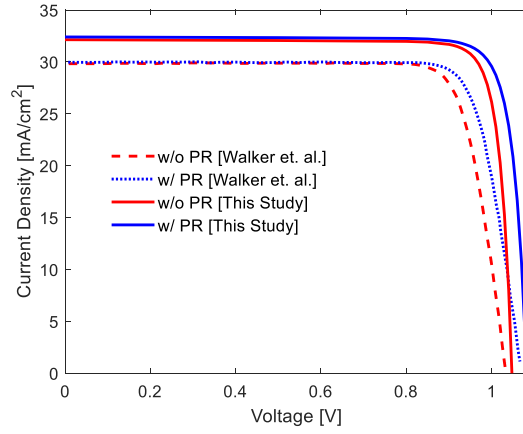


Figure 3.9: I-V curve simulated for thin-film GaAs solar cell using the photon-recycling model. The cell structure in this study is adopted from Walker et. al.

Table 3.4: Simulations results summary

Metric	Walker et. al.			This Study		
	Thin-Film Results [39]			Thin-Film Simulations		
	w/o PR	w/ PR	ΔV_{oc} [V]	w/o PR	w/ PR	ΔV_{oc} [V]
J_{sc} [mA/cm ²]	29.9	29.9		32.1	32.4	
V_{oc} [V]	1.033	1.069	0.036	1.048	1.082	0.034
FF [%]	82.1	82.4		86.8	85.8	
η [%]	25.3	26.3		29.2	30.0	

3.4.2 SELE and SCE of a thin-film GaAs solar cell

The SELE and SCE of this thin-film GaAs solar cell were then simulated and shown in 3.11. Figure 3.10 shows the SRH, radiative and Auger recombination rates (a); internal luminescence efficiency (ILE) (b); the energy band diagrams (c); and the internal electric field (d). Each figure shows the properties for short-circuit (SC), maximum power (MPP) and open-circuit (OC) working points. The subfigures (a) and (c) show the spatial distribution of each recombination mechanism and energy bands for each of the working points. The ILE and the internal electric field show the

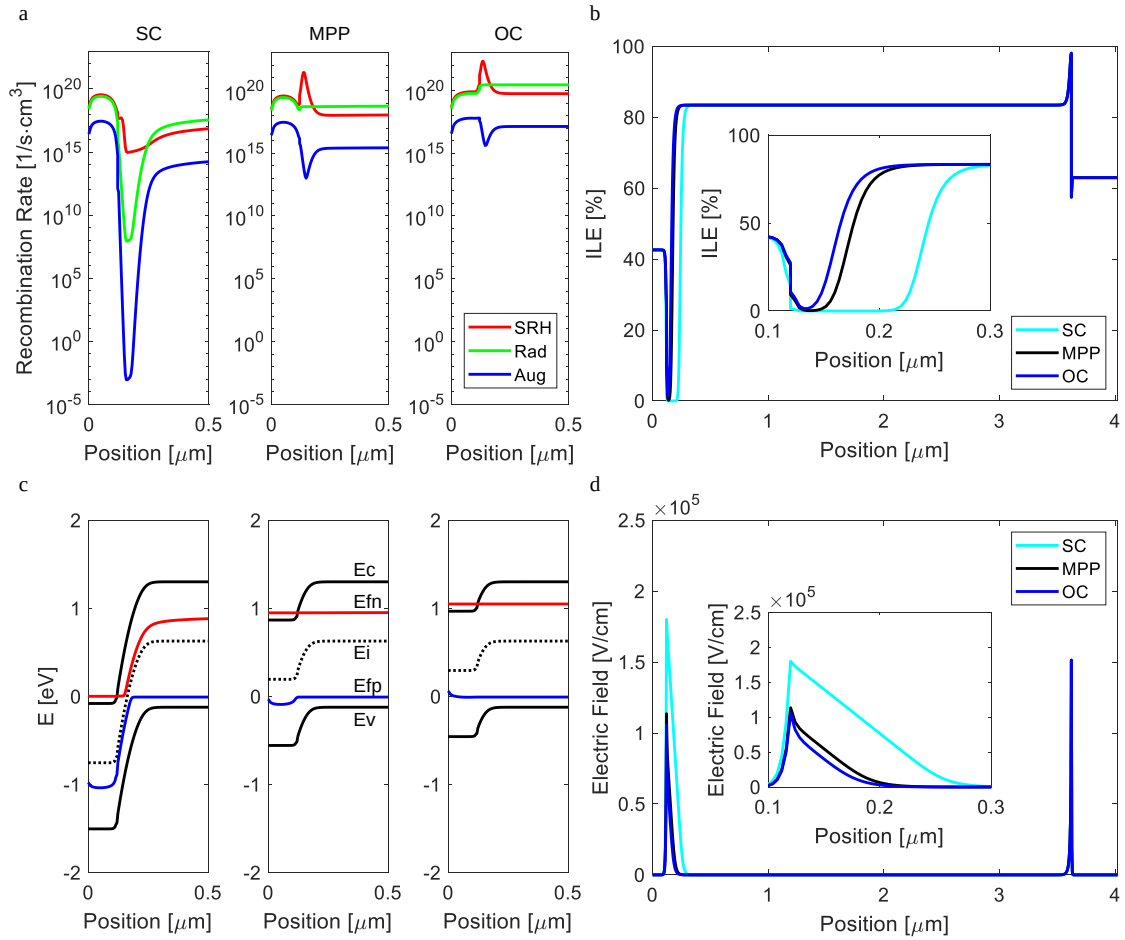


Figure 3.10: Simulation results of thin-film GaAs solar cell (0.12 μm emitter and 3.5 μm base). (a-b) different recombination rates and internal luminescence efficiency (ILE), (c-d) energy band diagram and internal electric field, under short-circuit (SC), maximum-power (MPP) and open-circuit (OC) working points. The inset in figures (b) and (d) shows the values for the ILE and the electric field around the junction

position of the junction. Under short circuit conditions, photo-generated mobile charge carriers are quickly swept away from the space charge region and are less likely to recombine, resulting in

locally low recombination rates as well as low ILE. The higher the operating point voltage is, the larger the build-up of the charge carrier in this region, resulting in higher recombination rates at and near the junction, and thus higher ILE near the junction.

The SELE and the SCE of this solar cell under short-circuit conditions can be seen 3.11. The SCE profile corresponds to the charge collection efficiency profile within the cell. For this profile, charges generated within the electric field of the cell will most likely be swept across the junction and collected, resulting in the highest SCE values at the junction. Further away from the junction, the SCE is reduced as a result of the diffusion length of the minority carriers in the material. Near the contacts, the ideal ohmic contact conditions in the simulation ensure a zero to no collection at the contacts, and thus zero SCE.

The SELE profile illustrates the competition between radiative and non-radiative recombination processes. As discussed in the previous sections, surface recombination reduces the SELE near the surface. Here, under ideal ohmic contacts conditions in simulation, no carrier buildup occurs near the contacts, which results in very low SELE at both ends of the device. However, in the depletion region, under short-circuit conditions, the electric field results in low carrier density and ILE. Carriers that are generated at the junction where high electric field is present, are more likely to be swept across and collected to current than to recombine radiatively, resulting in low SELE. Further away from the depletion region, the SELE increases. Charge carriers that are generated farther from the junction are less likely to reach it and are more likely to recombine radiatively. Thus, the slope of the SELE profile next to the surface or junction is in correspondence with the minority carriers diffusion length and the escape probability of the generated photons. In this sample, in the presence of a BSF layer and an internal electric field, charge buildup is diminished near the BSF layer which reduces the radiative recombination and thus the total SELE near the BSF region. Removing the BSF layer ($3.62\ \mu\text{m}$ to $4.02\ \mu\text{m}$) would have resulted in an SELE profile that is high at the middle of the base layer, and reducing to 0 at the contact.

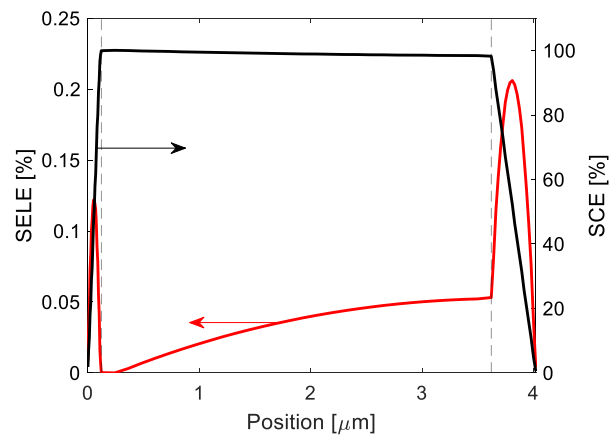


Figure 3.11: SELE (red) and SCE (black) simulations of a thin-film GaAs solar cell for short-circuit operating conditions. The SELE and the SCE profiles corresponds to positions where changes of the internal electric field occur within the device.

3.4.3 SELE and SCE under bias voltage

The light bias changes the $I - V$ characteristics of the cell. Therefore, when additional charge carriers, ΔG , are introduced in the simulation, the cell working point changes. For this reason, EQE and ELE measurements, as well as simulations for SELE and SCE, must be performed with a background illumination light bias as a baseline, so that any additional charge carriers can be considered a small perturbation.

Calculating the SELE under a voltage bias shows the effect of different areas on photovoltage build-up. For this, the SELE and SCE profiles as well as the ELE and EQE were simulated under different bias voltages. The SELE and SCE are calculated using equations similar to equations 1.8 and 1.9:

$$S_{ELE}(z, V) = \frac{\phi_{out}[G_0(z') + \Delta G \cdot \delta(z' - z), V] - \phi_{out}[G_0(z'), V]}{\int_z \Delta G \cdot \delta(z' - z)} \quad (3.2)$$

$$S_{CE}(z, V) = \frac{1}{q} \frac{J_{out}[G_0(z') + \Delta G \cdot \delta(z' - z), V] - J_{out}[G_0(z'), V]}{\int_z \Delta G \cdot \delta(z' - z)} \quad (3.3)$$

Here, G_0 is the optical generation profile resulting from background bias light, which was chosen here to be 1 sun. Thus, the output current or luminescence that originates from the applied bias voltage is deducted to obtain the effect of the photocurrent or photoluminescence only. The same can be applied for simulating the ELE and EQE:

$$\eta_{ext}(\lambda, V) = \frac{\phi_{out}[G_0 + G(\lambda), V] - \phi_{out}[G_0, V]}{\phi_{abs}(\lambda)} \quad (3.4)$$

$$EQE(\lambda, V) = \frac{1}{q} \frac{J_{out}[G_0 + G(\lambda), V] - J_{out}[G_0, V]}{\phi_{inc}(\lambda)} \quad (3.5)$$

Where $G(\lambda)$ is the optical generation from monochromatic light illumination at wavelength λ , and ϕ_{abs} and ϕ_{inc} are respectively the absorbed and incident photon flux that originates from monochromatic light only.

Figure 3.12a presents the simulated ELE results using equation 3.4 and Figure 3.12b shows the results using equation 3.5. The simulations are shown for bias voltages between zero and V_{oc} , and the insets show a magnification of the profile to distinguish the effect of the applied bias voltages on each profile. Corresponding to the ELE and EQE profiles, the SELE and SCE profiles are simulated using equations 3.2 and 3.3. Figures 3.12 c-d show the simulated SELE and SCE profiles for the same bias voltages. Figure 3.13 shows the simulated results of the internal device electric field and the quasi fermi level splitting.

A small variation in SELE and SCE is observed for the different applied bias voltages between short and open circuit conditions. As current collection is dominated by diffusion, changing the applied voltage hardly changes the collection efficiency. However, as seen in 3.13, in short circuit conditions, a stronger electric field and low quasi-fermi level splitting at the junction maintain

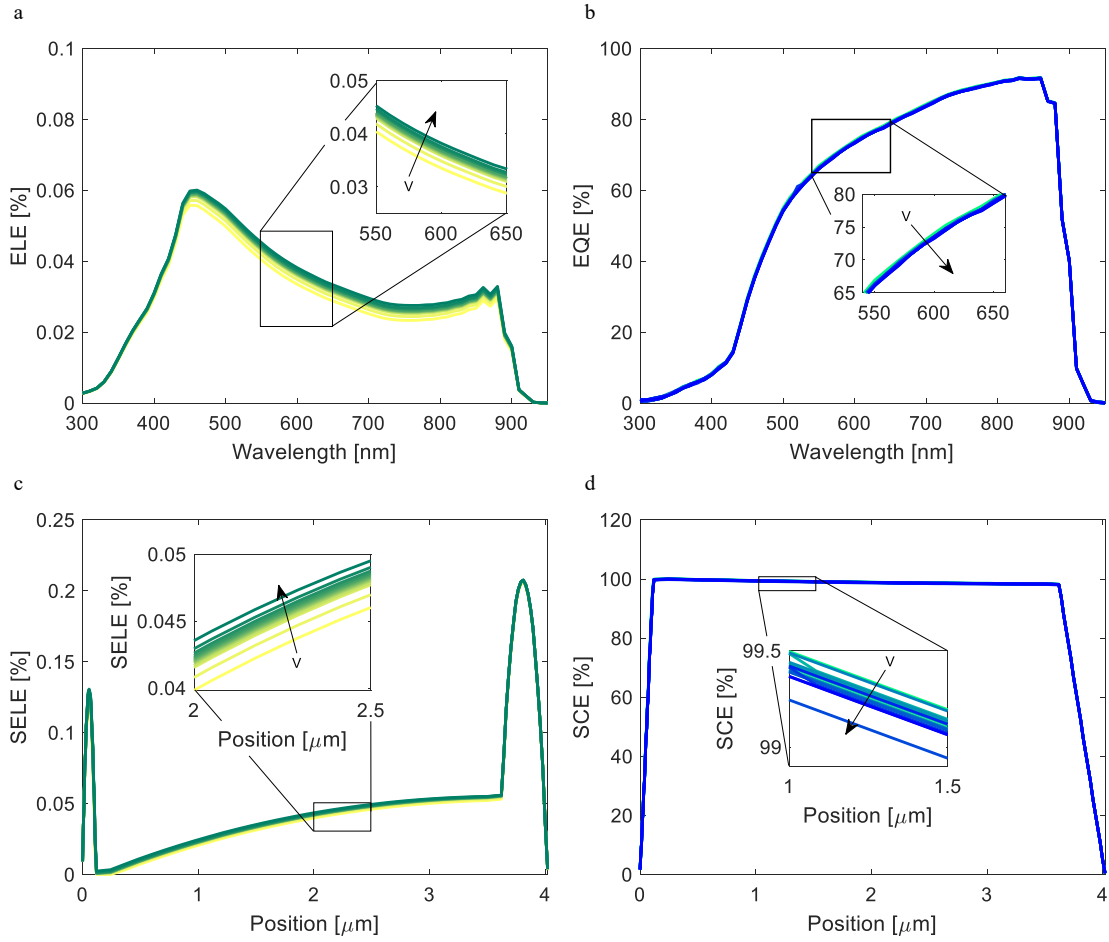


Figure 3.12: Simulated ELE and EQE profiles (a-b) and SELE and SCE (c-d) of a thin-film GaAs solar cell as a function of applied voltage bias on the device, V . Simulated ELE and EQE profiles were calculated using equations 3.5 and 3.4, and SELE and SCE profiles were calculated using equations 3.2 and 3.3. Insets show a magnification of the profile to distinguish the effect of bias voltage on each profile. The voltage span is between 0[V] and V_{oc} [V]

low carrier densities in the depletion region. This minimizes radiative recombination losses and results in slightly lower SELE and slightly higher SCE for low forward bias applied voltages. As voltage increases, the internal electric field is reduced, and carrier accumulation near the junction increases recombination, leading to a slight increase in the SELE profile. This phenomenon is most significant in the bulk of the device, where larger quasi fermi level splitting is observed.

To further validate the method, the ELE and EQE profiles are restored using the simulated SELE and SCE profiles and equations 1.7 and 1.3 and are shown in the results chapter appendix??.

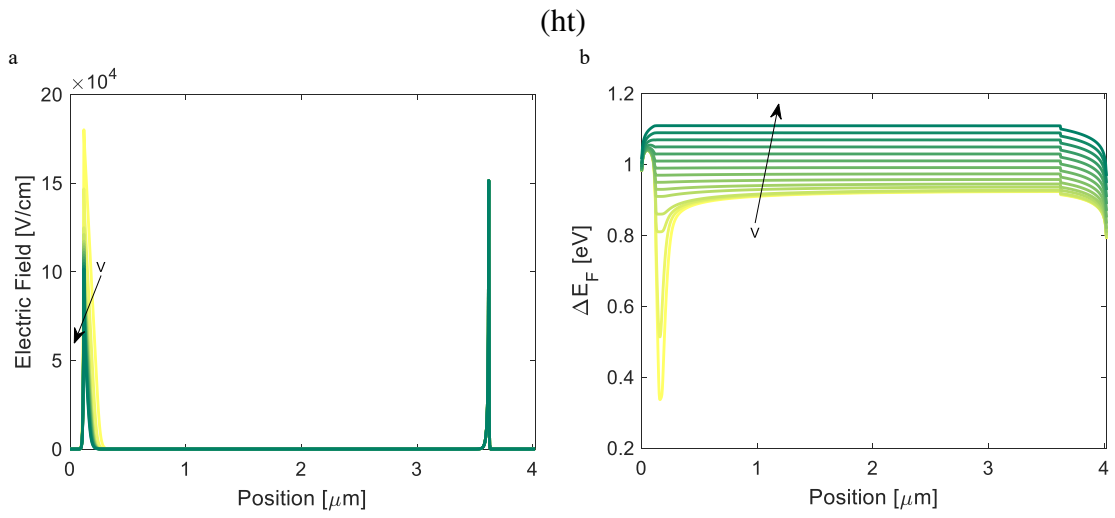


Figure 3.13: Simulated internal electric field (a) and quasi fermi level splitting (b) of a thin-film GaAs solar cell as a function of applied voltage bias on the device, V . The voltage span is between 0[V] and V_{oc} [V]. Changes in the electric field and quasi fermi level splitting explain the difference in SELE and SCE simulations for different voltage bias.

3.4.4 SELE simulations of thin-films solar cells of different materials

In this section, the SELE of thin-film solar cells with different materials was investigated. The solar cell model was simplified to a single PN junction structure for comparison. The SELE was simulated for GaAs, InP, and Si PN junctions. Each material was used in both the emitter and base. The optical generation profile was calculated using a simplified optical model that calculates the generation profile assuming no oxide layers or back reflection while under standard AM1.5G illumination. [56] The optical constants for GaAs and InP were taken from the ellipsometry measurements, while the optical constants for Si were taken from Schinke et al. [67]. The optical generation profile was then input for the COMSOL calculation. The material properties in the simulation are shown in table 3.5. The dimensions and doping were set to match the previous thin-film solar cell structure of GaAs as seen in section 3.4.1.

Table 3.5: Various Solar cell material properties used in simulation

Parameter	GaAs		InP		Si	
	n-type	p-type	n-type	p-type	n-type	p-type
Doping [cm^{-3}]	$2 \cdot 10^{18}$	$9 \cdot 10^{16}$	$2 \cdot 10^{18}$	$9 \cdot 10^{16}$	$2 \cdot 10^{18}$	$9 \cdot 10^{16}$
Thickness [μm]	0.12	3.5	0.12	3.5	0.12	3.5
Mobility [cm^2/Vs] (holes, electrons)	(70, 2290)[23]	(280, 4500)[23]	(50, 1900)[23]	(130, 3500)[23]	(110, 190)[68]	(340, 800)[68]
Radiative recombination coefficient [cm^3/s]	$1.87 \cdot 10^{-10}$		$\approx 1.0 \cdot 10^{-10}$ [57]		$\approx 5.0 \cdot 10^{-13}$ [69, 70]	
Auger recombination coefficient [cm^6/s] (holes, electrons)	$1 \cdot 10^{-30}$		$9 \cdot 10^{-31}$ [58]		$(9.9 \cdot 10^{-32}, 2.8 \cdot 10^{-31})$ [71]	
SRH lifetime [ns] (holes, electrons)	(2, 300)	(2, 300)	(150, 300)[57, 59]	(150, 300)[57, 59]	(1, 1)[μs] [72, 73]	(1, 1)[μs] [72, 73]

SELE and SCE simulation results are shown in figure 3.14. The low external luminescence of Si, as a result of its indirect band-gap, is clear compared to the GaAs and InP SELE results. The SELE differences is in the same order of magnitude as of the radiative recombination coefficient of each material. The higher the radiative recombination coefficient, the higher the probability of photogeneration at some position to contribute to the PL of the device. In GaAs and InP, most of the SELE originates from the emitter. This is due to the predominance of the radiative recombination process in these materials. For the Si cell, the majority of the SELE is located in the base, where most of the absorption and emission occur. Additionally, in comparison with the GaAs thin-film cell from the previous section, the SELE shown here is lower near the back surface. This is due to the effect of the back surface field. The back surface field layer pushes carriers from the back contact, and created a higher charge density at the interface. This increases the likelihood that radiative recombination occurs within the BSF layer.

The SCE simulation of all of the materials showed no practical differences. As expected, the SCE is close to one near the space-charge region and decreases away from the junction due to the ideal ohmic contact at the back surface.

Using equation (3.1), the spatial contribution to the photo-voltage (the term in the integral) and the photo-voltage loss for different device thicknesses, $\Delta V_{oc}(d)$, are shown in figure 3.15. Figure 3.15b shows that almost the entire photo-voltage is built within the first $1\mu m$ for GaAs and

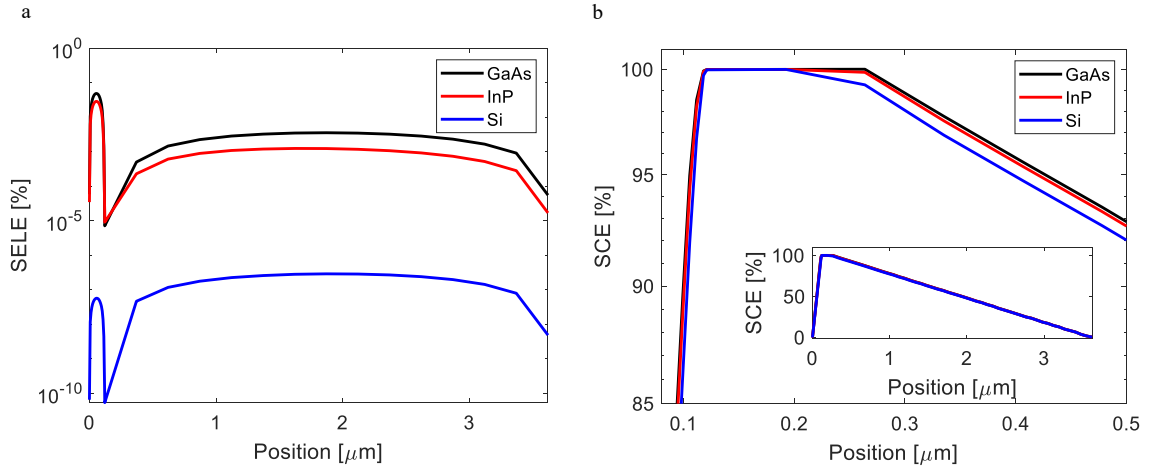


Figure 3.14: Simulated SELE profiles of thin-film solar cells using different materials. The solar cell model is a single PN junction structure with dimensions and doping that are described in the previous section 3.4.1. The SELE was simulated for GaAs, InP, and Si PN junctions. Each material was used in both the emitter and base.

InP cells. This suggests that in terms of the obtainable photo-voltage, the absorber layers can be much thinner for cells made of these materials. However, for Si cells, the analysis suggests that the open-circuit voltage increases with thickness and is more sensitive to the absorber layer thickness. This analysis shows how improving the SELE of the device improves the photo-voltage losses. The SELE can therefore be used as a tool for studying the effect of different regions within the device on the output voltage.

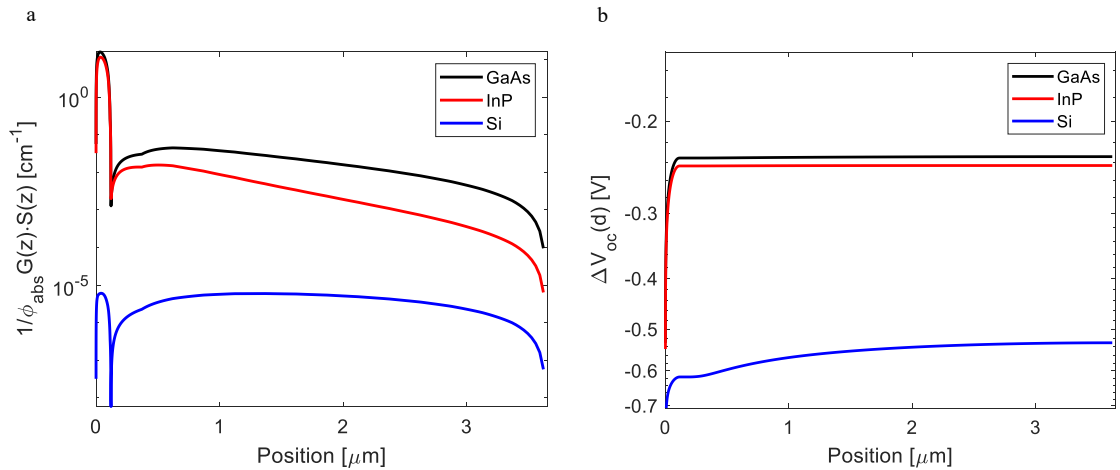


Figure 3.15: (a) The spatial contribution to the photo-voltage for thin-film PN junction solar cells made of different materials, calculated by using equation (3.1), (the term in the integral). (b) The photo-voltage loss, $\Delta V_{\text{oc}}(d)$, for different device thicknesses, d , calculated by using equation (3.1).

3.5 SELE Surface Characterization Using MOS Capacitors

Studying the SELE in MOS capacitors allows to investigate the effects of surface recombination and electric field dynamics in a controlled device. The MOS capacitor offers a platform on which the surface charge and electric field above a substrate can be changed by controlling the voltage on its interface. In this section, we present the results of our SELE simulations and measurements, which were performed over a MOS capacitor that was made with the GaAs wafer discussed previously as the substrate of the device. This section shows how the SELE can be used to study variations in surface states and internal electric fields within a device.

3.5.1 MOS Capacitor Simulation Model

In this section, the SELE analysis is implemented on a MOS capacitor that is simulated in COMSOL. The modeled MOS capacitor was comprised of p-type GaAs as a substrate as well as a transparent conductive contacts as the gate. This MOS capacitor structure allows to analyze the SELE profile for different surface charge distributions in both accumulation and depletion modes by applying different bias voltages.

The MOS capacitor device was simulated using the "thin oxide layer" feature in COMSOL. The relative permittivity of the oxide was taken to be 9, which corresponds to an Al_2O_3 oxide layer. The oxide thickness was taken to be 100nm and the top contact work function was taken to be 4.6eV which is that of ITO. The parameters used in the MOS capacitor simulation are shown in table 3.6. The material properties were based on the GaAs substrate used in the previous sections. Some parameters were slightly changed to allow the convergence of the simulation in multiple voltage working points. In this simulation, the "PARADISO" solver achieved faster results than the default "MUMPS" solver when running many simulations for each bias voltage and was therefore used.

Capacitor Characterization

Capacitance- voltage (C-V) measurements are used to show the transition between the accumulation, depletion and inversion modes. The device C-V characteristics are simulated by conducting a DC voltage sweep with an additional small harmonic perturbation that is considered a small AC signal. Figure 3.16 shows the simulated capacitance for high and low frequency frequency perturbations. This result shows that the simulated capacitor is in accumulation for voltages under 0[V]. Depletion occur for voltages between 0 and 10[V] and inversion for voltages above 10[V].

Parameter	GaAs MOSCAP	Referenced Value
Doping [cm^{-3}]	$5 \cdot 10^{17}$	-
Thickness [μm]	350	-
Mobility [cm^2/Vs] (majority, minority)	(180,4500)	[23]
Radiative recombination coefficient [cm^6/s]	$2 \cdot 10^{-10}$	$1 \cdot 10^{-10}$ [49]
Auger recombination coefficient [cm^6/s]	$1 \cdot 10^{-30}$	$10 \cdot 10^{-30}$ [50]
SRH lifetime [ns] (holes, electrons)	(1, 1)	(0.82, 1.9) [51]
Surface Recombination velocity [cm/s]	$4 \cdot 10^5$	$2 \cdot 10^6$ [52, 74, 54]
ITO Gate Thickness [nm]	100	-
Oxide Thickness [nm]	100	-
Oxide Relative Permittivity	9	rajab et al
Top Contact Work Function [eV]	4.6	Nehate et al

Table 3.6: GaAs MOS capacitor material properties used in simulation

3.5.2 SELE simulations of a MOS capacitor

In MOS capacitors, different bias voltages affect the carrier concentration at the semiconductor-oxide interface. In a p-type GaAs substrate, applying a positive voltage on the isolated gate attracts mobile electrons towards the surface, as seen in figure 3.17 shows a diagram of a GaAs MOS capacitor. This effect is similar to an increase in surface recombination velocity, whereas when a negative voltage is applied, mobile electrons are pulled away from the surface, which is similar to a decrease in surface recombination velocity. In this way, extracting the SELE while applying bias voltage to a MOS capacitor allows to study the SELE profiles for different surface properties. SELE simulations of GaAs MOS capacitors were performed following equation 1.8. The optical generation was modeled using the TMM as described in the methods section 2.2.3. The effect of the bias voltage on the PL signal can be seen in figure 3.18. This simulation shows how the PL emitted flux is increasing by almost eight folds from the depletion mode to accumulation. Figure 3.19 (a-b) shows the simulated ELE and SELE profiles for different bias voltages accordingly. The increase in emitted photon flux aligns with the increase in the total ELE, as seen in Figure 3.19a. Figure 3.19b shows the SELE profiles for each applied voltage. For this device, the SELE changes mostly near the surface, which corresponds to changes in carrier concentration in this region. Under positive voltage bias, mobile electrons are attracted towards the surface. These charges are then more likely to recombine non-radiatively at the surface, which implies a lower PL signal and lower SELE in that region, similarly to what occurs with large surface recombination velocity. However, when a negative voltage is applied, mobile electrons are pulled away from the surface, increasing the depletion layer. Charges are then less likely to recombine non-radiatively, which implies higher PL and SELE, similarly to what occurs with low surface recombination velocity. This result is therefore similar to the SELE simulation result of a simple GaAs wafer that is performed with varying surface recombination velocities, as seen in figure 3.3.

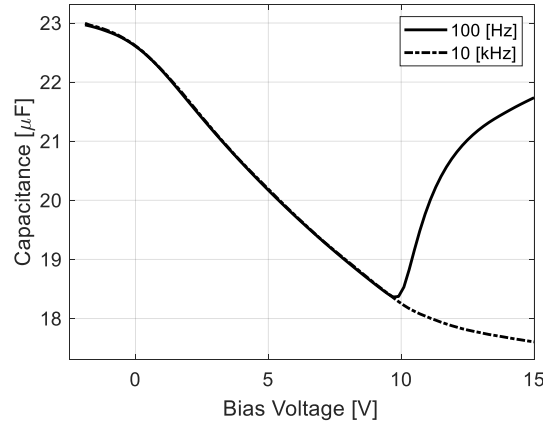


Figure 3.16: C-V measurement simulation using different AC frequencies. C-V measurements are used to identify the accumulation region in a capacitor and is used here to identify that the model behaves as a capacitor

Interestingly, the ELE profiles in the accumulation and depletion modes show different trends. As seen in 3.19a, under negative voltage bias, the ELE increases more for shorter wavelengths, resulting in a more uniform, flattened ELE profile. This points out to the emission probability deeper within the surface, as can also be seen in the SELE profile. The carrier generation profile is the same for each simulation. However, as charge carriers are pushed from the surface for these voltages, higher carrier concentration is created deeper within the device. This case is similar to when carriers are generated deeper within the device, due to absorption by higher incident wavelengths. Thus, for this case the ELE profile is higher in the 500-600 nm incident wavelength range.

The SELE characterization of MOS capacitors suggests the possible application of this method on similar devices. Consider, for example a hypothetical MOSFET with GaAs substrate and an ITO transparent gate. In this case, a detailed SELE map can be extracted by using a highly focused supercontinuum laser that is rastered across the channel width. Figure 3.21 shows a hypothetical result of the SELE of a MOSFET device at pinch-off. In this example, the variation of charge carriers near the substrate-oxide can imply a change in photon emission, and thus SELE along the channel. Thus, this SELE mapping can be used to identify some of the channel properties, such as the depth of the channel or perhaps defects along its length. Defects along the channel length reduce the total current flow and degrade the transistor quality. Such defects are likely to locally reduce radiative recombination and thus the SELE at that position. Therefore, mapping of SELE around crucial points within a device can be used to create a characterization tool, while providing an in situ and non-destructive characterizing method.

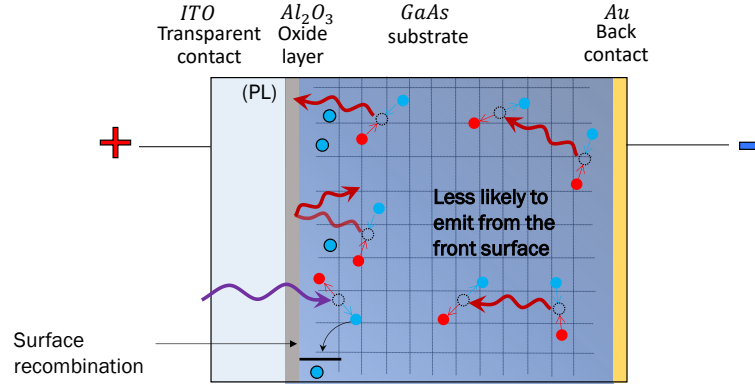


Figure 3.17: MOS capacitor diagram under bias voltage. The capacitor is designed with conductive and transparent ITO contacts to allow illumination and emission of PL from the substrate. Under applied voltage, carriers can either attract or repel from the contacts, creating a depletion region near the substrate and oxide interface. Recombination at this interface affect the total PL emission greatly than deep within the substrate

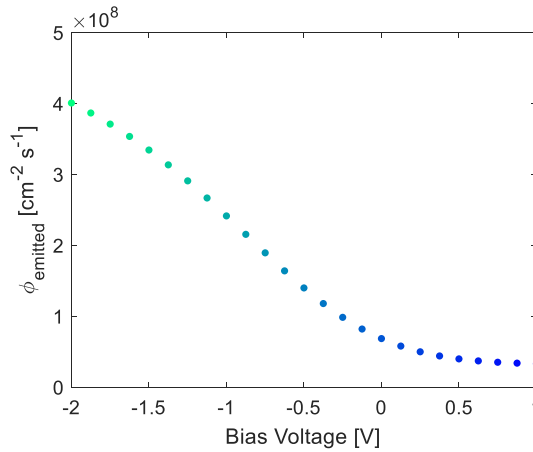


Figure 3.18: Emitted PL photo flux from MOS capacitor simulations. In negative applied bias voltages the capacitor is in accumulation mode, reducing the amount of minority charge carriers at the surface, effectively reducing the surface recombination rate and increasing the total emitted photon flux

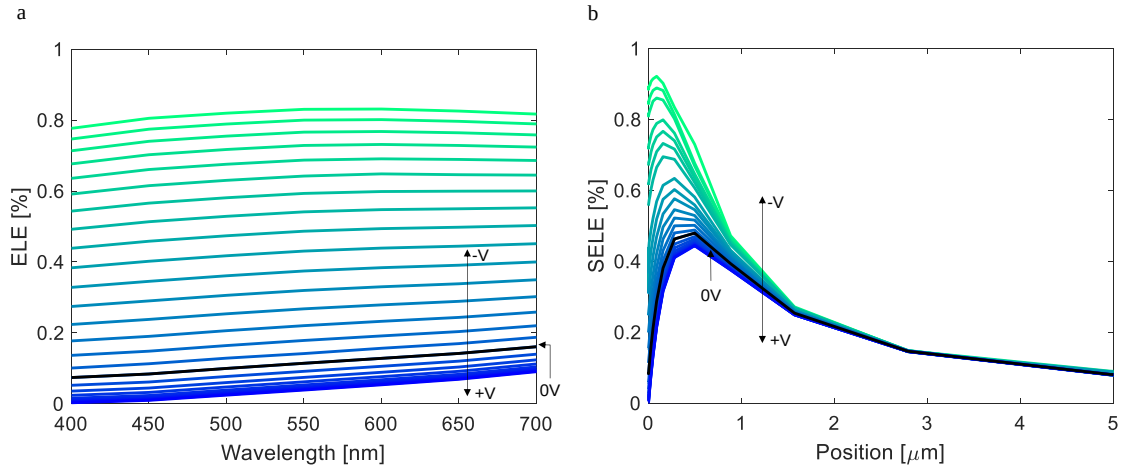


Figure 3.19: ELE and SELE profiles of a p-type GaAs MOS capacitor in simulation. (a) The incident wavelength-dependent ELE simulation results for different bias voltage from -2V to 1V. (b) Simulated SELE profiles for each bias voltage. The black lines correspond to the simulation at 0V.

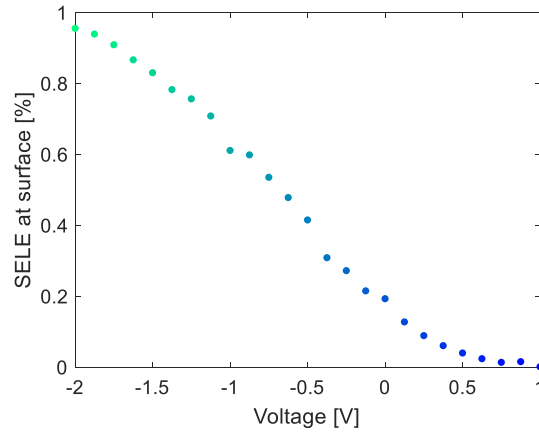


Figure 3.20: SELE values at the surface for MOS capacitor simulations. In negative applied bias voltages the capacitor is in depletion mode, reducing the amount of charge carriers at the surface, effectively reducing the surface recombination rate and increasing probability for emission for charge carriers that are generated near the surface, which corresponds to a higher SELE

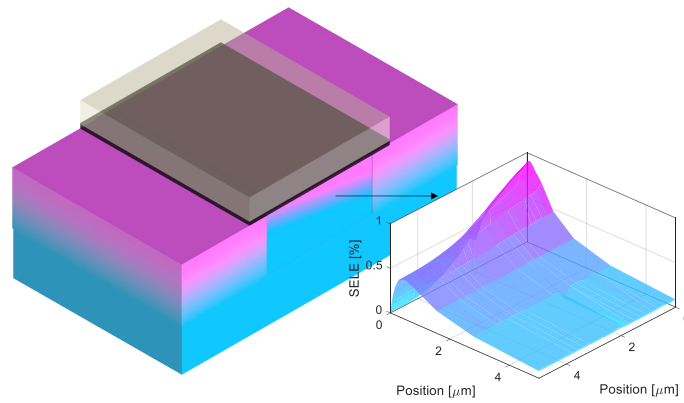


Figure 3.21: Hypothetical example of SELE implementation on MOSFET device. The results for the MOS capacitor are shown as an example for possible results for MOSFET SELE experiment, where a change of depletion depth along the channel correspond to a different profile of the SELE, as shown. For such experiment, one should have accurate illumination source together with a transparent and conductive gate material on the MOSFET device

3.5.3 SELE extraction from measurements of a MOS Capacitor

Measurements were performed on a MOS capacitor device fabricated with 40 nm Al_2O_3 as the isolation layer and 180 nm ITO as the gate. The substrate used was a double side polished, $(1.74 \pm 0.83) \cdot 10^{19} cm^{-3}$ Zinc-doped, 350 μm thick GaAs wafer (Wafer Technology Ltd.). The fabrication process was carried out on a 1x1 cm wafer sample which was then diced into small 2x2 mm samples. Surface pretreatment was performed on the substrate as discussed in the methods section. However, only ITO annealing was performed. The thickness of the ITO and oxide layers was obtained by fitting the measured and calculated reflectivity of the sample, as shown in figure 3.22. The reflectivity measurement was performed using a UV-VIS spectrophotometer. TMM was used to model the reflectivity, where the optical constants were obtained from ellipsometry measurements. The SELE extraction was performed only on the GaAs layer, and was assumed to be 0 anywhere else. The SELE was extracted using the wide-range Tikhonov regularization method, as described in the methods section. 2.4.2. The regularization was performed using the $L0$ operator and a linear mesh from 0 to 1.5 μm . The sample substrate is assumed to be homogeneous. Therefore, the SELE profile is expected to be similar to the SELE extracted of a simple wafer. Thus, for the MOS capacitor sample, only SELE profiles that exhibit one maximum only were taken in account. Any profiles with more pronounced peaks would indicate some erroneous solutions and were therefore excluded.

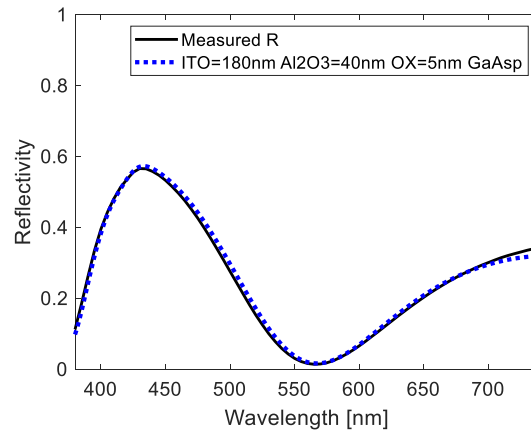


Figure 3.22: Reflectivity measurements of GaAs MOS capacitor together with TMM reflectivity calculations. The layer thicknesses in the device was determined using a fit to the measured reflectance and the result for each layer is shown in the legend

Electroluminescence (EL) can also contribute to the measured luminescence signal, particularly for defective devices with some charge trapping and junction behavior. To avoid overestimating the external luminescence due to EL effects, a correction was applied. For each voltage, the

luminescence of the device was measured without external illumination and subtracted from the PL measurements. This baseline, corresponding to the EL contribution, was determined separately and removed to isolate the true photo-excited external luminescence. The following results are presented using this correction.

The ELE and extracted SELE profiles that were obtained under various applied voltages are shown in figure 3.23 (a-b) respectively. The ELE and SELE profiles increase under negative voltage and decrease slightly for positive voltage, which is consistent with simulation in the previous section. 3.5.2 The change in the ELE is more pronounced for shorter incident wavelengths, indicating a change in surface properties that occurred under the electrical bias. These variations correspond to the large change in SELE profile near the surface, as seen in figure 3.24. However, applying a negative bias results in a higher SELE near the surface, as predicted by the device simulations in figure 3.20. Similarly, as voltage increases, charge carries are attracted to the surface, recombine non-radiatively, and the SELE at the surface is reduced. This experimental demonstration shows that the SELE can be used as a characterization method to study and analyze surface recombination and recombination dynamics at interfaces within a device.

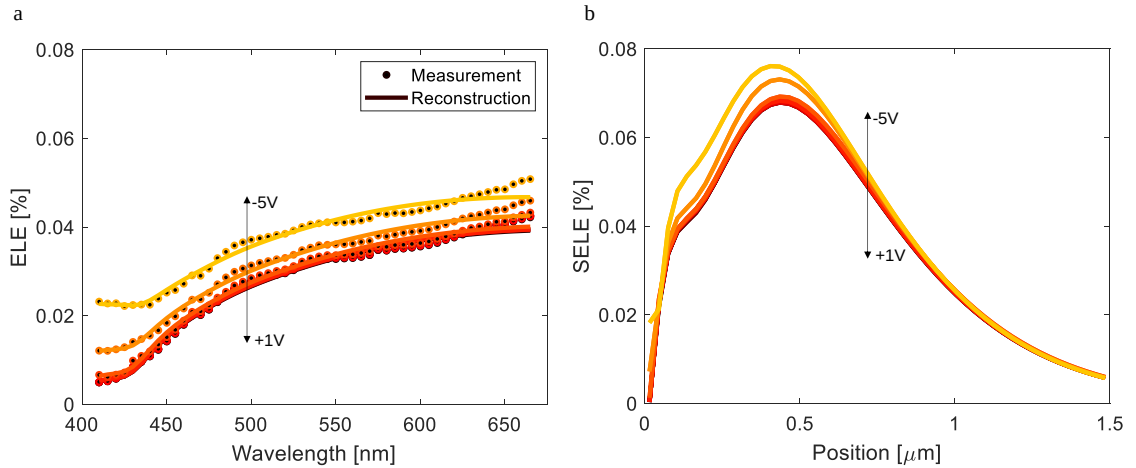


Figure 3.23: ELE and SELE extracted profiles of a p-type GaAs MOS capacitor. (a) The incident wavelength-dependent ELE measurements and reconstruction of results from the SELE extracted profile, results are shown for bias voltage from -5V to 1V. (b) Extracted SELE profiles for each bias voltage.

After the measurement and due to the higher applied voltage, the above sample experienced some electrical breakdown and it was not possible to repeat the measurements. Thus, more samples and measurements are to be conducted.

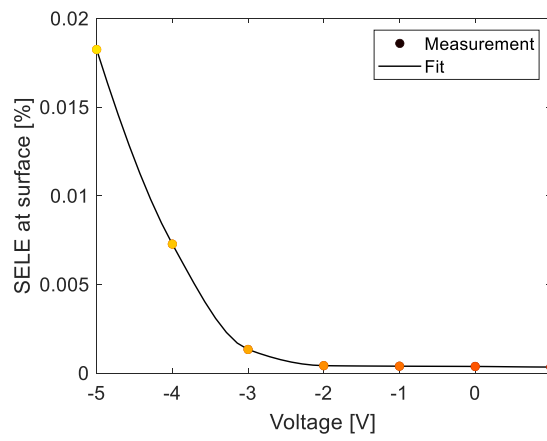


Figure 3.24: The SELE at the semiconductor-oxide interface of the GaAs MOS capacitor as a function of the applied voltage. The values were taken from figure 3.23. The solid line represents a fitted function from the resulted values.

3.6 Operando Dynamic SELE of Native Oxide Formation

The oxide layers of a material have a significant affect on the recombination at the surface. Usually, an exposed surface rapidly forms a layer of native oxides. This layer acts as a barrier, passivating the available surface states. Thus, during the formation of this layer, the surface recombination velocity and thus the PL change. This section describes the dynamic SELE profile during the formation of the native oxide on a GaAs p-type wafer. The SELE is measured over time and compared to the oxide layer thickness formation.

3.6.1 Sample Preparation

The sample investigated in this section was a p-type GaAs wafer (100) that underwent chemical etching prior to PL measurements using a combination of alkaline base (NH_4OH) and hydrochloric acid (HCl) to study the effect of native oxide formation on PL and SELE. The substrate used was a double side polished, $(1.98 \pm 0.28) \cdot 10^{19} cm^{-3}$ Zinc-doped, $350\mu m$ thick GaAs wafer (Wafer Technology Ltd.). The sample preparation and removal of the native oxide layer is described in the Methods section. 2.8.3 After the removal of the native oxide layer, the sample was transferred to the PL measurement setup for measurements.

3.6.2 Dynamic SELE of Native Oxide Formation

PL and ELE measurements were performed immediately after etching to assess the impact of surface treatment on the SELE. In this example, the developing of an oxide layer occupy surface states and reduces the carrier recombination at the surface, and the surface recombination velocity. Therefore, at the beginning of the measurement, when a low-to-no oxide layer is present, low PL emission is observed. As the layer forms, the less charge carriers recombine at the surface, leading to a slightly higher PL. Correspondingly, since the SELE at the surface is sensitive for surface recombination, the SELE at the surface increases with time. Figure 3.25 shows the ELE and SELE change over time, obtained from a sample that underwent chemical etching. Figure 3.26 shows the SELE at the surface change over time, together with the oxide layer thickness following Lukas' model. [2]. In this figure, the SELE increases over time and follows a logarithmic increase rate, that is similar to the oxide layer time. This figure shows that the SELE at the surface can be compared to the oxide layer thickness, suggesting that this method can be used to measure dynamic processes that are related to rapid change in surface properties for a variety of semiconductor devices.

To improve the results in this section, a further experiment is suggested. An identical sample can be cleaned and etched and then transformed to an ellipsometer where multiple measurements would be taken at constants time intervals. This analysis would enable to determine the oxide thick-

ness for each time step accurately, thus obtaining better results for each of the ELE measurements and further accurate the precision of the SELE profile.

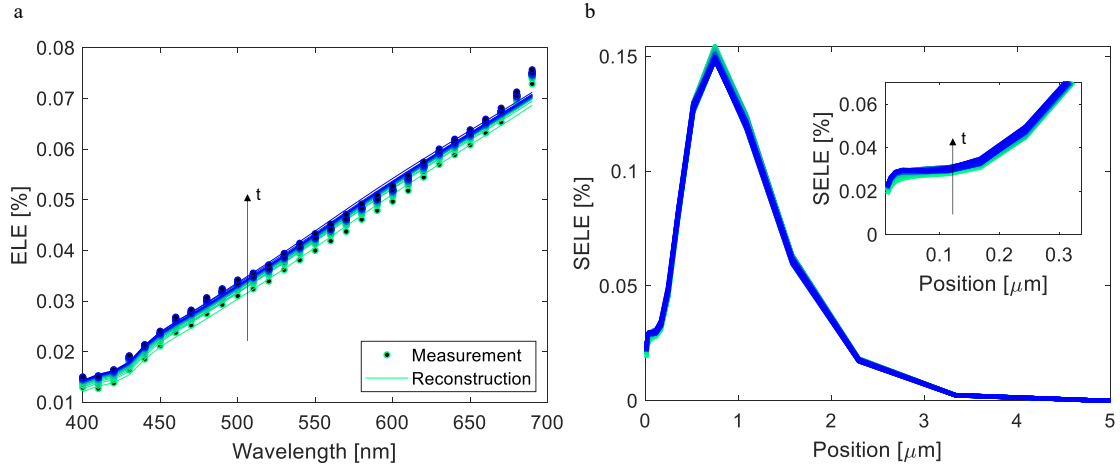


Figure 3.25: ELE and SELE profiles of a p-type GaAs wafer samples over time, t , following chemical etching. (a) The incident wavelength-dependent ELE measurements and reconstruction of results from the SELE extracted profile (b) Extracted SELE profiles corresponding to the ELE measurements over time. Inset shows the SELE profiles near the surface. The measurements were taken between 3 minutes and up to 1 hour and 45 minutes, with 3 minutes intervals.

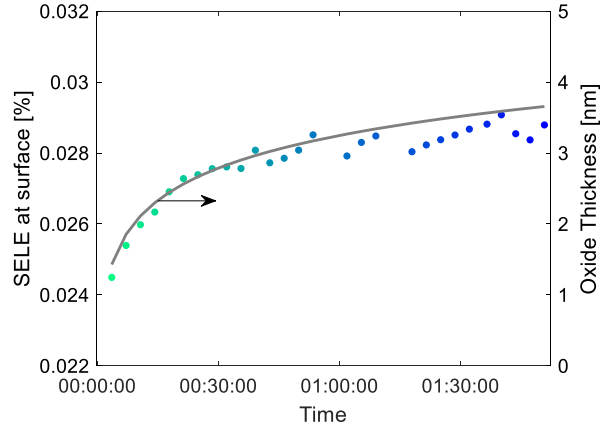


Figure 3.26: The SELE at the surface of an etched p-type GaAs over time. Etching was performed with a combination of $(\text{NH}_4)\text{OH}$ and HCl etching. The SELE is compared to the oxide thickness formation model (solid line).[2]

3.7 Results Summary

The study began with the initial investigation of SELE of a simple p-type GaAs wafer. Experimental SELE profiles were acquired and compared with simulations to show key features of the SELE profile. The agreement between the experimental data and the simulations validated the SELE extraction method. The SELE profile of p-type GaAs showed the effect of surface recombination together with the bulk lifetime of the material. Using device simulations, it was possible to fit the material, surface, and diffusion properties to those of the measured sample.

The SELE can thus be used to measure surface properties, such as surface recombination velocities, based on the SELE values at the surface. This analysis was also used to calculate the open-circuit voltage loss for a device with the sample's material. The voltage buildup was calculated for different device thicknesses using the SELE result. This analysis is used to obtain a suggested thickness for a device that achieves lowest photo-voltage loss under 1 sun illumination.

Following the SELE of a p-GaAs wafer, SELE of a p-InP wafer was investigated and compared with GaAs under similar conditions. The SELE results for InP reveal its high carrier mobility and lifetime, demonstrated by the SELE peak position at $4.5\mu m$ (vs. $1.5\mu m$ in GaAs) and the energy band simulations, which show deeper carrier penetration and higher quasi-fermi level separation. These properties allow carriers to avoid surface recombination, reducing front-surface PL. InP also shows significantly lower ELE values than GaAs, probably due to its higher carrier lifetime, surface recombination velocities, and higher doping in GaAs, as well as differences in back-surface reflectivity (non-specular).

Next, SELE simulations of PN junctions were shown to understand the SELE characteristics in solar cell junction configurations. The simulations reveal that the SELE can identify junction locations as it decreases in regions with strong internal electric fields. The electric field present in junctions, or in the interface between different layers, sweeps charge carriers and reduces local radiative recombination, thus affecting the SELE profile at that location. The SELE profiles, analogous to SCE analysis, are shown along with the simulated recombination rates, ILE, and band diagrams for SC, MPP, and OC conditions to show the SELE change for each working point.

The third part of the study focused on the characterization of SELE with respect to surface recombination. MOS capacitors were simulated to explore the behavior of the SELE in a GaAs MOS capacitor device under different voltages. The SELE profiles provided information on the interaction between the electric field and the carrier dynamics within the device. Applying a bias voltage on the capacitor can attract or retract the carriers from the surface. These carriers then fill or deplete the surface states to increase or reduce the recombination of the surface. The affect is shown in the SELE close to the surface. In a p-type capacitor, for example, a negative voltage pushes electrons away from the surface, expanding the depletion region and reducing surface recombination, thereby enhancing PL and SELE.

Simulations show that PL emission can increase by up to eightfold between the accumulation and depletion modes. SELE changes occur primarily near the surface, indicating that the bias voltage modulates the surface recombination dynamics. Experimental results from MOS capacitors made with an ITO transparent gate confirm simulation trends. The measured ELE and SELE profiles change with voltage, especially under short-wavelength illumination, reinforcing the fact that the applied voltage affects mainly surface recombination. A hypothetical GaAs MOSFET example is shown, suggesting that SELE could serve as a non-destructive, in situ diagnostic tool to probe channel properties in sub-micrometer devices.

Lastly, the SELE of the native oxide formation of p-GaAs was shown. An p-type GaAs is etched using *HCl* to remove any native oxide layer. Immediately after etching, the sample's PL is measured over time, and the SELE profile is extracted for each measurement. This result showed that the SELE at the surface changes along with the oxide layer thickness formation. Suggesting that the SELE can be used to investigate a dynamic process effect on surface properties and the SELE potential as a tool in situ measurement.

4 Conclusions and Future Work

This work introduced and demonstrated a method for characterizing depth resolved external luminescence efficiency (SELE) using PL measurements and optical modeling. This method can be used to gain deeper insights into the optical and electrical performance of semiconductor materials and devices. Using SELE analysis, the depth-resolved information on loss mechanisms, self-absorption processes, and the contribution of different regions to the photovoltage from a light absorber devices or solar cells is presented. The strength of this method lies in its operando ability to probe surface and interface properties and in-depth defects across a sample, offering valuable information that cannot be captured through conventional measurements. SELE serves as a powerful indicator of local recombination dynamics and carrier extraction efficiency, which makes it particularly relevant for photovoltaic and optoelectronic applications.

In this work, the method was first demonstrated using a simple GaAs wafer. The well-understood structure and properties of GaAs allowed to establish the SELE extraction method using PL measurements and available optical modeling together with finite element simulations. This step provided a platform for the development of the experimental setup, measurement procedures, and data analysis. Simulations were used throughout the work to guide the interpretation of the PL and SELE data and to validate the physical intuition behind the observed features. Here, SELE simulations showed the potential photovoltage gain of GaAs based devices for different passivation properties.

Using the SELE extraction method, the SELE of a simple InP wafer was shown and compared to the SELE results of GaAs. The SELE profiles, results, and simulations are used to compare the different material and surface properties.

The SELE profiles of PN junctions were then studied, demonstrating the SELE profiles of functioning PV cells. In this analysis, the SELE profile was found to map the location of internal electric fields as well as the space charge region within a device. In addition, the SELE profiles of PV cells made of different materials and their potential photovoltage loss were shown. In this section, the SELE was used to map both the photovoltage losses of functioning devices and to suggest optimal device thickness to minimize voltage loss.

The SELE near the surface was found to be highly sensitive to surface recombination, while

the internal luminescence efficiency and the self-absorption of photons determine the SELE further away from the surface. To investigate the SELE near the surface, the SELE of a transparent gate MOS capacitor with a GaAs substrate was analyzed. The MOS capacitor structure allowed to control the carrier density near the surface by applying an external bias voltage. This change in carrier density near the surface affects the surface recombination and thus the SELE. The SELE was used to quantify the performance of surface properties.

The last part of the results showed the *operando* and dynamic SELE of a sample that underwent the formation of a native oxide layer. A GaAs sample was etched and transferred to the PL measurement setup, where PL measurements were taken over time. Corresponding to the oxide layer thickness, the SELE near the surface changed over time, showing the SELE capabilities for characterizing a dynamic and fast passivation process.

This method directly complements other *operando* methods, such as spatial collection efficiency (SCE) analysis, which maps charge transport mechanisms. The SELE analysis, incorporated with current measurements and SCE extraction, is suggested for future implementations in PN junctions. An integrated analysis can help mapping both non-radiative recombination centers and current collection capabilities within a device. Thus, affectively creating a depth resolved map of all losses within a device. Therefore, the SELE can provide a key step towards rapid optimization of materials and interfaces for photovoltaic cells.

The measurement methodology can be further expanded. One approach involves the use of a broadband super-continuum laser source, which would allow probing a sample at different excitation wavelengths with high localized beam and power. This would enable spectral selectivity in spatial mapping, opening the door to investigating more complex multilayer structures. Future capabilities of this method has its potential for high-resolution 3D mapping of surface recombination or in-depth defect points using SELE analysis. This can be achieved by raster scanning the excitation beam across the sample surface. Such mapping can enable the extraction of information and the distribution of luminescence and recombination centers within a device.

Applying this method to full device structures, including functioning transistors or PV cells could provide insight into how local fabrication variations or defects influence performance at the microscale. Such a capability would be invaluable both in research and in industrial quality control, where identifying and localizing fault points early can drive significant improvements in device reliability and efficiency.

The SELE provides a flexible experimental platform for exploring spatially resolved recombination centers and surface properties of devices and materials. The combination with device simulation using COMSOL, allowed to validate the results and identify key features in SELE profiles. The SELE provides a valuable tool for studying semiconductor materials, surface properties, devices, and dynamic processes within a device. With future implementations and development, including integration with spectrally tunable sources, advanced rastering techniques, and combi-

nation with current collection and SCE analysis, this method holds promise for a wide range of applications in next-generation opto-electronics.

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A n-InP PL Instability Over Time

A.1 *PL signal Instability*

This appendix presents a series of PL experiments conducted on n-type InP samples as part of the initial investigation for this study.

n-type Sulphur InP (100) samples (WaferTech) with $1.35 \cdot 10^{17} [cm^{-3}]$ carrier concentration were tested with the goal to evaluate the suitability for further measurements and method development. However, the PL response of the InP samples exhibited instability over time. With constant illumination power and measurement conditions, the emission intensity was changing significantly over large periods of time. Figure A.1 shows an example for PL emitted flux measurements over 4 hours. In this experiment, the lamp was stabilized for 20 min prior illumination on the sample. The experiment started by opening an external shutter and can be seen in this figure by the large increase of PL signal over a short period of time at the 7 seconds mark. This time-dependent behavior was consistently observed across multiple experiments, including sequences where the sample was illuminated, allowed to rest in the dark, and then re-illuminated, as shown in A.2. Each cycle showed a variation in the PL signal, indicating a lack of stability that prevented the use of these samples for quantitative analysis.

Despite finding that these samples might be unsuitable for the the method development, the observed PL fluctuations may be linked to some surface states, charge trapping, or other dynamic processes at the material interface. To test this possibility, the samples were subjected to a brief hydrofluoric acid (HF) etch in an attempt to remove surface oxides or passivated surface states. However, this treatment did not lead to any change in the PL stability. These preliminary findings suggested that while the n-type InP samples could not be used in the development of the SELE method. An improved measurement method with fast PL measurement might be a useful for future investigations into time-dependent PL and SELE.

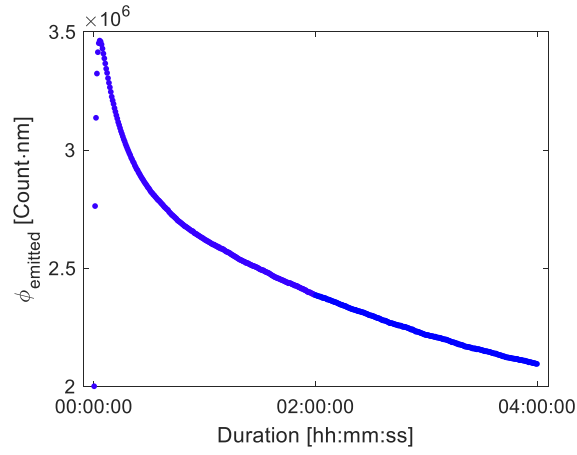


Figure A.1: PL emission of n type InP wafer sample under constant illumination over time. The illumination power was measured prior to the experiment at 1.5 mW

A.1.1 Temperature Dependence

PL measurements were also taken while measuring the samples temperature. In this setup, a thermocouple was connected to the back side of the sample while illuminating the front side on the integrating sphere setup. However, no significant temperature change was observed, while PL signal was changes similarly to what presented in figure A.1.

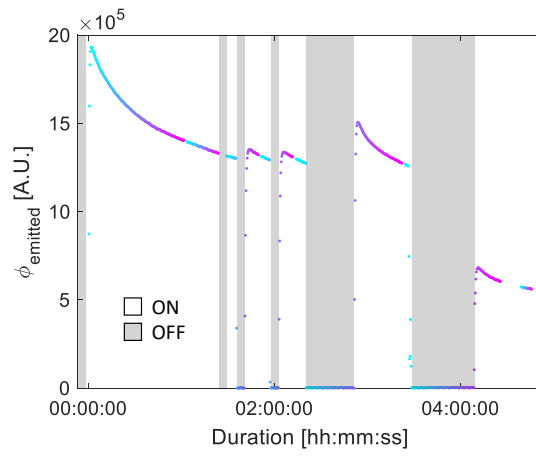


Figure A.2: PL emission of n type InP wafer sample for different illumination periods. The gray background represents the time where the illumination was off.

B Optical Constants Details

B.1 Verification of optical constants

B.1.1 *R and T measurements and calculation comparison*

To verify the extracted optical constants, the transmission and reflection spectra were calculated using an optical simulation and compared with measurements. The simulated transmission and reflection spectra were calculated using GenPro4 software. [38] For this calculation, the optical constants and thicknesses of each layer were set as input. Figure B.1b shows the simulated and measured optical transmission and reflection.

B.1.2 *PL simulations and calculation comparison*

Special attention is required to calculate the spectral PL emission using extracted optical constants from ellipsometry measurements. [35] The PL simulated spectra (calculation method described in the methods section 2.6) was used to further verify the extinction coefficient k and the calculated absorption coefficient α . The inset in figure B.1a shows the resulted absorption edge near 880 nm which is consistent with the high free carrier density of the sample [75]. The absorption of sub-band-gap photons through free carrier absorption (FCA) results in intra-band transitions that do not lead to the generation of mobile charge carriers⁵. Hence, the spontaneous photon emission and ultimately the PL spectrum are unaffected by this process. The Drude term in the optical model describes the contribution of free carriers to the dielectric function of metallic materials [76, 77]. Thus, to properly calculate the PL emission spectrum, the optical constants of the sample were extracted from the same GenOsc ellipsometry model, excluding the contributions from FCA (i.e., not including the contribution of the Drude term). The difference in the optical constants can be seen in the inset in Figure S2.1a. The measured PL spectrum, and the spectrum simulated based on the modified optical constants and equation (10) are shown in figure B.2. The incident wavelength is 600 nm. The reflection mode configuration refers to an experiment (and simulation) in which only PL photons that are emitted from the surface on which the input illumination is incident

are accounted for. In the transmission mode configuration, only photons emitted off the opposite surface are considered.

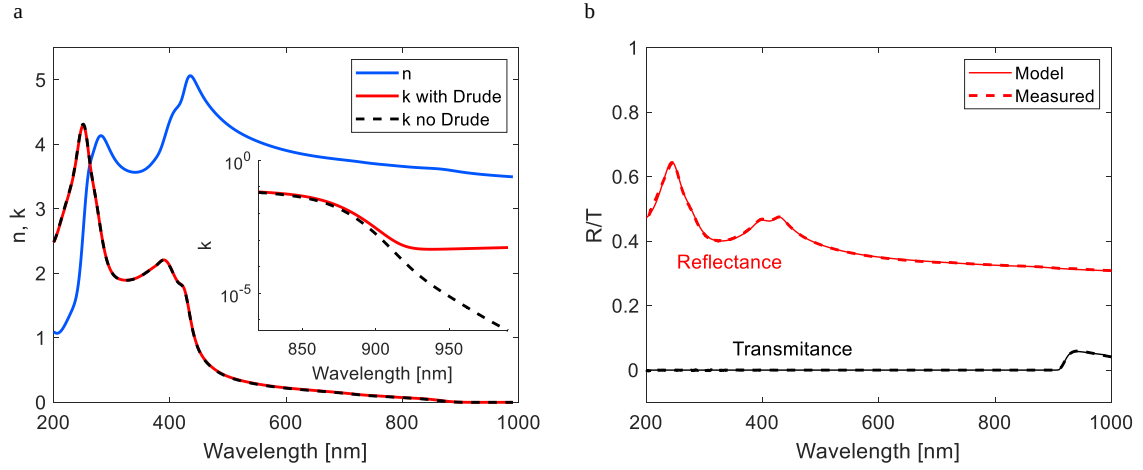


Figure B.1: (a) Optical constants of the GaAs wafer obtained from the ellipsometry analysis with and without the Drude oscillator which accounts for free carrier absorption. The inset shows the extinction coefficient focusing on wavelengths near the absorption edge. (b) Modeled and measured reflectance and transmittance.

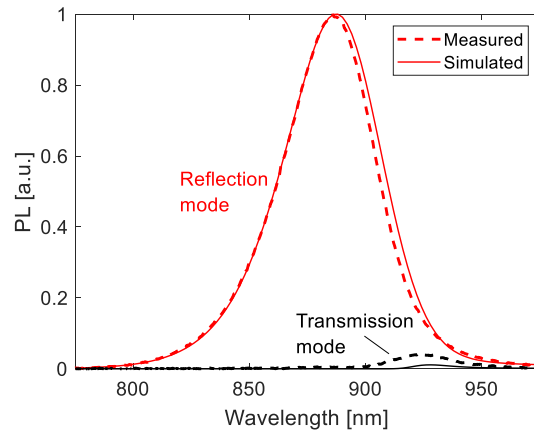


Figure B.2: Simulated and measured PL spectra in reflection (when collecting photons emitted from the surface on which the input illumination is incident) and transmission mode (when collecting photons emitted from the surface opposite to the surface on which the input illumination is incident). Both measured spectra were normalized by the maximum value of the measured reflection-mode PL spectrum, and both simulated spectra were normalized by the maximum value of the simulated reflection-mode PL spectrum. The incident wavelength is 600 nm.

C Optical Model Details

C.1 Emission Angle and Photon Escape Probabilities

To accurately simulate PL emission while accounting for all optical emission angles, reflection and transmission calculations were performed for photon energies around the band-gap. Reflection and transmission calculations were performed using the transfer matrix method (TMM). In addition to reflection and transmission, we evaluated the mean escape and absorption probabilities by averaging all photon energies, following the approach outlined by Steiner et al. as shown in C.1. These results provide a comprehensive view of the PL emission from GaAs wafers, which helped to design the free space measurement setup, as described in the methods section. 2.1.4

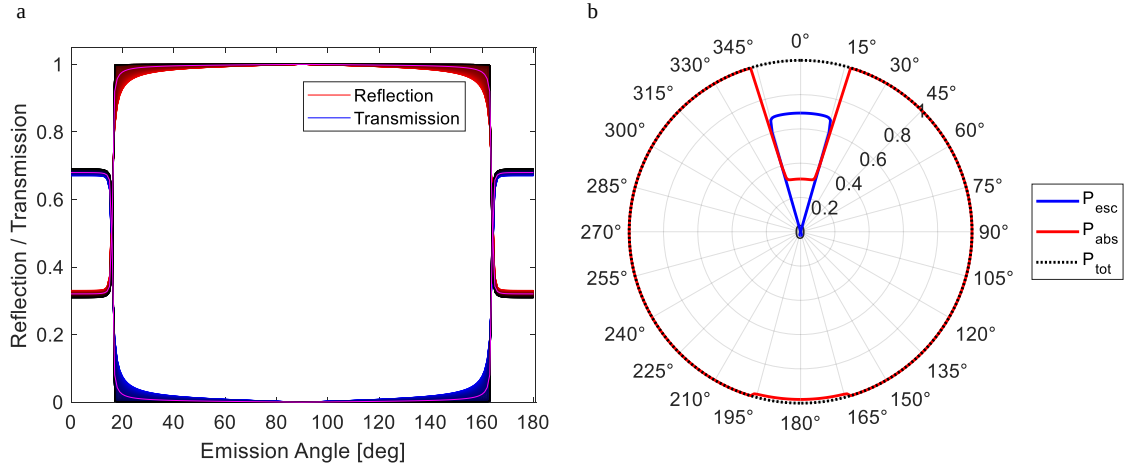


Figure C.1: (a) Reflection and transmission TMM calculations for GaAs wafer, each shade represents a different photon energy. The magenta line represents the photon energy that corresponds to the GaAs band gap. (b) Mean escape and absorption probabilities calculated over all photon energies. Probabilities calculations followed Steiner et al.

C.2 Photon Recycling Model Validation

Figure C.2 shows a comparison between the wavelength-dependent photon escape and absorption probabilities, averaged over the thickness of the sample, and the escape and absorption probabilities, calculated using the model presented by Steiner et al. [9]. In this comparison, free carrier absorption was not considered. This agreement between the two calculations confirms the validity of the presented photon recycling model.

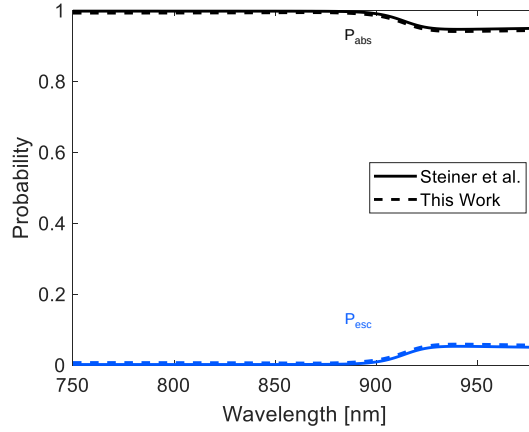


Figure C.2: Photon absorption and escape probabilities calculations, $P_{abs}(E)$, $P_{esc}(E)$ using the photon recycling model (presented in this work) and following Steiner et al. 11. The probabilities are averaged over the sample thickness.

C.3 PR Effect on Band diagram and Emission Spectrum

Figure S4.4a shows the band diagram of a p-type GaAs wafer when photon recycling effects are considered and when they are not. Photon recycling dramatically increases the quasi-Fermi level separation deep within the material. Figure S4.4b shows the simulated reflection and transmission (inset) PL spectrum with and without the iterative photon recycling calculation. As expected, the introduction of photon recycling effects increases the PL intensity. However, since the escape probability deep within the material is small, the large quasi-Fermi level separation induced by photon recycling does not contribute as much to the PL intensity in reflection mode.

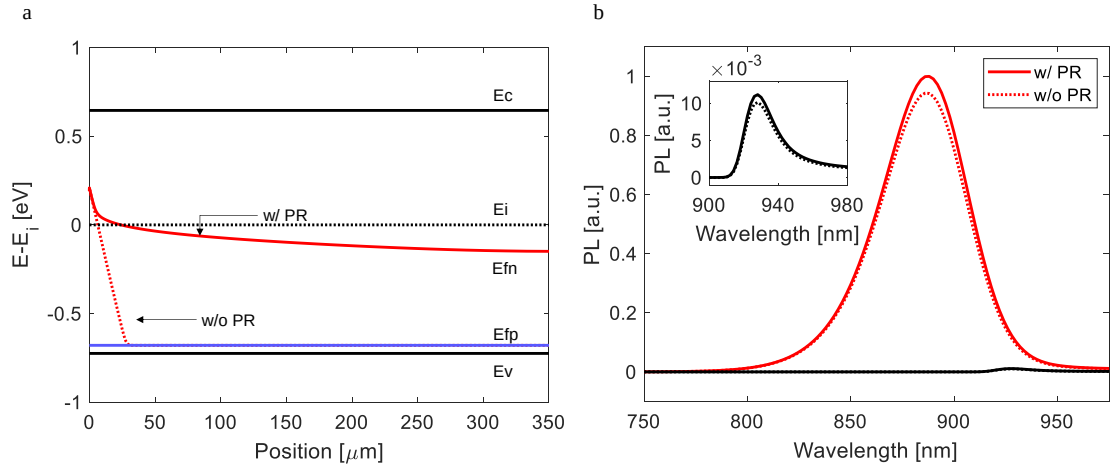


Figure C.3: (a) Energy band diagram with and without the inclusion of photon recycling effects. (b) Simulated PL spectrum (normalized) with and without the inclusion of photon recycling effects. Inset shows the PL spectrum in transmission mode. The sample properties are detailed in Table S5.1. and the incident wavelength is 600 nm

D Instruments Evaluation

D.1 Light Source Output Power Stability

To ensure the accuracy of the optical measurements, the stability of the power source was evaluated prior to its use. The output power was tested over time until a stabilized signal was achieved. The power output was observed to stabilize only after a warm-up period. Based on this time, a waiting time of 15 minutes (900 seconds) was introduced before any measurements were made. The results of the stability tests is shown in figure below.

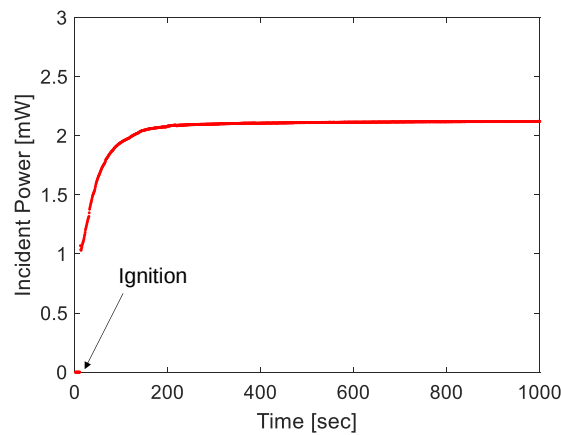


Figure D.1: The illumination power output over time, measured prior to PL measurements. As a result of this measurement, a stabilization time of at least 15 minutes was taken prior to any PL measurements

D.2 Optical Fiber Evaluation

D.3 Spectrometer Readings Validation

To ensure the reliability of the spectral data analysis, the readings of the spectrometer were cross-validated between the manufacturer's software and the MATLAB acquisition. A p-type GaAs sample was measured using multiple integration time while using the smallest entrance slit for the spectrometer to detect small differences in the spectrum. The results are shown in figure D.2. Both measurements showed identical results using multiple integration times. The results confirmed that the MATLAB interface accurately reflects the spectrometer's output. This validation step provides that the spectrometer data acquired and analyzed via MATLAB throughout the thesis is consistent with the instrument's intended performance.

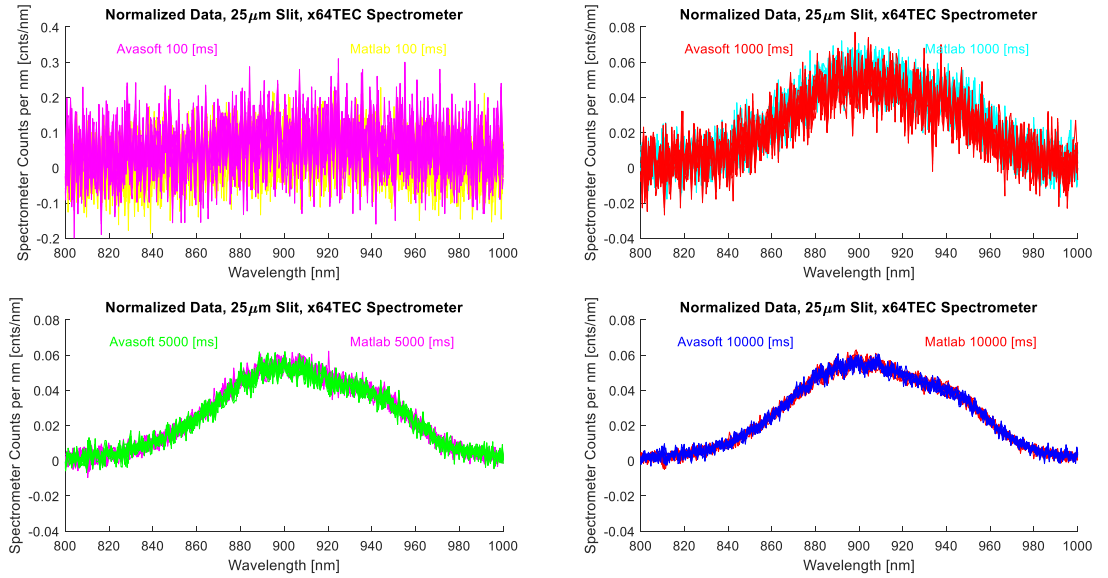


Figure D.2: Matlab vs Avantes PL signal comparison for different spectrometer slits and integration times. This measurement was used to ensure that the data recorded in Matlab using the spectrometer instrument commands is exactly similar to the data that is recorded when using the official spectrometer software.

D.4 Spectrometer Sensor Temperature

The spectrometer used in this work includes a thermoelectric cooler (TEC) to regulate the sensor temperature, which is critical to minimizing noise in low-light measurements such as photoluminescence. The manufacturer recommends operating the sensor at a maximum $\Delta T = 20^\circ\text{C}$ below

ambient, typically around 5C, to achieve stable performance. However, a sensor temperature of 0C resulted in some lower noise levels. Based on this observation, all measurements were performed with the sensor stabilized at 0C, ensuring a improved signal-to-noise ratio even for low integration times and weak PL signal.

D.5 Error Analysis

D.5.1 PL signal linearity with Incident Power

To be able to obtain faint signals, the spectrometer was tested for non-linearity with the measurement integration time. The linearity was tested using PL measurements of GaAs p-type wafer using multiple measurements and different integration times. The PL area multiplier showed a clear linear behavior over the different integration times.

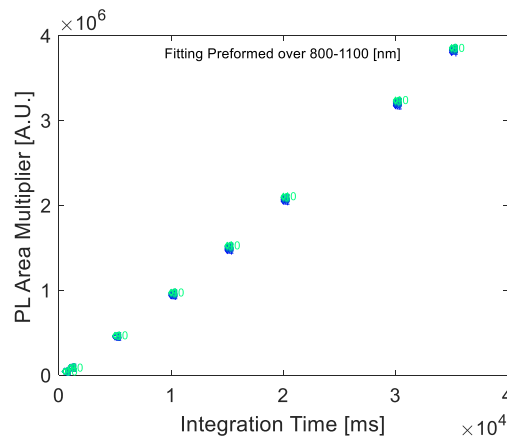


Figure D.3: PL emitted multiplying factor for different spectrometer integration times (i.t.). The first PL emission for 10 seconds i.t. is set as 1. For each i.t., 10 PL measurements are taken, and are shown with the measurement number. This result was used to verify the linearity of measured PL for different integration times of the spectrometer.

D.5.2 Power meter and Photo-diode Measurement Error

Accurate optical power measurements are essential for the reliability of ELE data. Therefore, the performance of the power-meter used throughout this work was evaluated. Attention is required to a calibration instrument, as small deviations can introduce significant errors in absolute power readings. Compared with a calibrated photo-diode, the power meter was found to have an overall

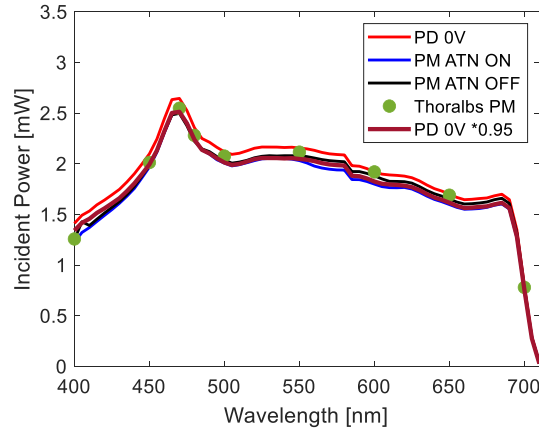


Figure D.4: Incident illumination power measurements for every illumination wavelength, using different power meters. This measurement was used for calibration between measurements taken with each instrument. (PD) Thorlabs Calibrated photodiode FDS1010-CAL under constant 0V, along with the results using a 0.95 factor. (Thorlabs PM) S175C detector with PM100USB power meter. (PM ATN ON/OFF) Newport 818-UV/DB detector with 1936-R power meter.

uncertainty of approximately 5% throughout the incident wavelength range, as shown in Figure D.4. This implies that the reported ELE can be considered accurate to within 5%, providing a reasonable confidence level for all optical measurements discussed in this thesis.

D.6 PL correction methods

D.6.1 PL measurements with Lock-In

The lock-in technique serves as a phase-sensitive detection method that can enhance the signal-to-noise ratio in PL measurements by selectively extracting signals that are modulated at a reference frequency using an optical chopper. The resulting luminescence signal therefore inherits this modulation pattern. The lock-in amplifier detects the PL signal with the reference modulation, effectively filtering out noise contributions. This approach permits detection of PL signals with orders of magnitude weaker than the background noise, which is particularly beneficial when measuring samples with low PL quantum efficiency. Using a Lock-In device for future detection of PL emission can increase the sensitivity of the measurement system, allow small PL signals to be measured, and improve the resolution of the system. This can be useful especially for the detection of fast SELE and PL dynamics which changes over time, such as n-type InP as shown in Appendix A. This will allow low integration times to be used and exciting the sample for less time, which can be crucial for such dynamic measurements.

E Integration Sphere Setup Radiance Calibration

Radiance calibration of the integrating sphere setup was attempted using an intensity calibrated lamp (Bentham CL2 RI Spectral Radiant Intensity Standard) that is used together with a stabilized power supply (Bentham 610 Current Stabilized Lamp Power Supply). The lamp intensity is calibrated for each emitting wavelength and is used to calibrate the measurement setup of the sphere, fiber and spectrometer. The lamp was placed at 50 cm from the sample port opening of the sphere, on the line of PL emission. Figures E.1 and E.2 show pictures from the calibration attempt in the lab. After a stabilization period of 15 min, a spectrum was recorded by the spectrometer and the spectrometer reading are calibrated according to the lamp's calibrated intensity specification.

However, this calibration method was found to be inconsistent with the calibration method mentioned above. Some of the reasons could originate from the large port opening compared to that of the sample PL emission spot. The PL emission is generated only at the incident illumination spot which was 0.16 cm^2 , where the port opening simulates light emission from a much larger area. Additionally, the light from the lamp enters the sphere with some irradiance cone that is different than that of the sample's emission. For these reasons the calibration of the measurement setup was performed as mentioned above in section 2.1.2.

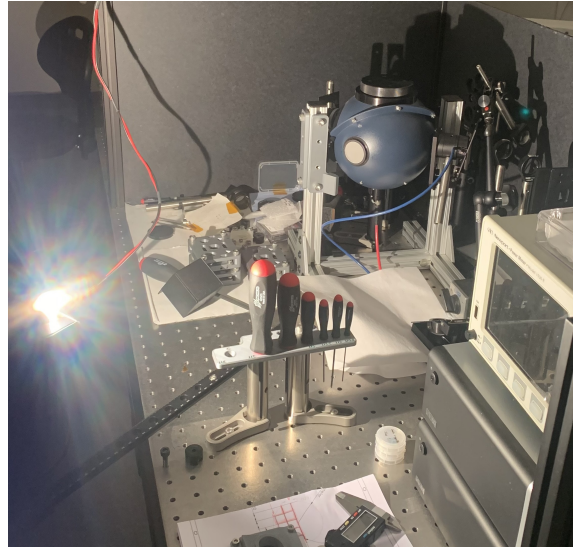


Figure E.1: Radiance calibration of the Integrating sphere setup using an intensity calibrated lamp. The lamp is situated at 0.5m from the sample port opening of the sphere



Figure E.2: Picture of the power supply recording of the current and power during the calibration using an intensity calibrated lamp

F MOSCAP SOP Document

As part of the process of fabricating a GaAs MOS capacitor device, a detailed Standard Operating Procedure (SOP) document was comprised and sent to labs in different universities with better fabrication capabilities. Attached here is the SOP document for reference purposes.



School of Electrical Engineering, Tel-Aviv University
Energy Devices Lab



GaAs MOS capacitor with ITO transparent gate

ITO- Al_2O_3 -GaAs(p)-Ti/Pt/Au

Requirements (SOP)

Contact: Tamir Yeshurun (tamir1@mail.tau.ac.il)

1

1. General Requirements:

- We wish to produce a GaAs capacitor with a transparent gate.
- The purpose of this process is to build a device which is a capacitor, so that light is able to illuminate through the front transparent gate, therefore the oxide layer should be quite thin so it will be transparent.
- The sample should be a capacitor. Therefore, the layers should be non-porous homogenous layers, with no "leaks" and no short contact.
- Sample size is not defined; however, we prefer half of a 2" wafer, or a 2.5 cm x 2.5 cm sample.
- Substrates are provided.

2. The Following is a suggested procedure based on a paper by Yang et al.[1]:

2.1. Substrate: 350 μm thick samples of n-type or p-type GaAs substrates with doping of $\sim 2 \times 10^{18} \text{ cm}^{-3}$ (samples supplied by us)
Possibly single 2" wafer that will be cut post process.

2.2. Surface pretreatment[2]: NH_4OH based surface pretreatment. Soak the samples in (NH_4OH) (29%) solution for 3 min to remove native oxide and rinse in flowing deionized (DI) water. Gently dry the surface using an N_2 blowgun.

The (NH_4OH) etching step is supposed to remove arsenic and gallium oxides from the surface.

2.3. Al_2O_3 ALD deposition: 20nm ALD Al_2O_3 layer on top side only, at 300 °C using precursors of TMA and water.

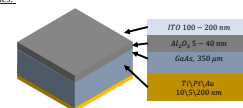
2.4. N_2 Annealing: 30s at 600°C RTP in N_2 ambient.

2.5. ITO, E-beam evaporator: 100nm ITO evaporation as the gate electrodes. ITO mixture: typically, 90% In_2O_3 , 10% SnO_2 by weight (according to the paper).

The feature of this layer is to be used as a transparent gate for this device creating a MOS capacitor.

2.6. Ti/Pt/Au, E-beam evaporator: Ti-10nm, Pt-5nm, Au-200nm to create electrical contact on the back of the substrate.

3. Schematics:



2

Figure F.1: GaAs MOS capacitor SOP document

G Miscellaneous

G.1 Experiment Automation

The measurement process was fully automated using Matlab instrument control. Custom scripts were developed to control and synchronize the operation of the instruments, allowing for improved repeatability. These automation routines helped maintain consistent measurement conditions and significantly reduced the time required for complex or repeated measurements.

Scripts for Matlab instrument control are available for the following instruments: HORIBA iHR320 monochromator, HORIBA filter-wheel (using robot mouse click macros with AutoHotkey software), Avantes spectrometer control, Newport 1936-R power meter, Thorlabs PM100USB power meter, Mightex LED, KEITHLEY 2450 source measure unit (I-V or photodiode measurements), HP/Agilent 4284a LCR meter. The readings from the Avantes spectrometer were compared to readings from the matlab control code. This validation can be seen in D.3

G.2 Holders

G.2.1 Sample holders

To ensure repeatable and accurate measurements throughout this work, several custom sample holders were designed and fabricated using a 3D printer or CNC machining. These holders were tailored to the requirements of each measurement setup, taking into account factors such as optical alignment, electrical contact stability, and sample placement. By securing the samples in a consistent orientation, these holders helped to ensure the reliability and reproducibility of the experimental data. The following section provides an overview of the different sample holders used, including their design considerations and practical implementation in the measurement setups.

Wafer samples

Accurate positioning holder for $350\mu\text{m}$ 2" half-wafers used to measure PL signal from a specific spot on the wafer due to concerns regarding different PL signal results from different locations on the wafer. This sample holder was 3D printed in PLA and was designed to fit half of 2" wafers, aligning the sample using the wafer's flat.

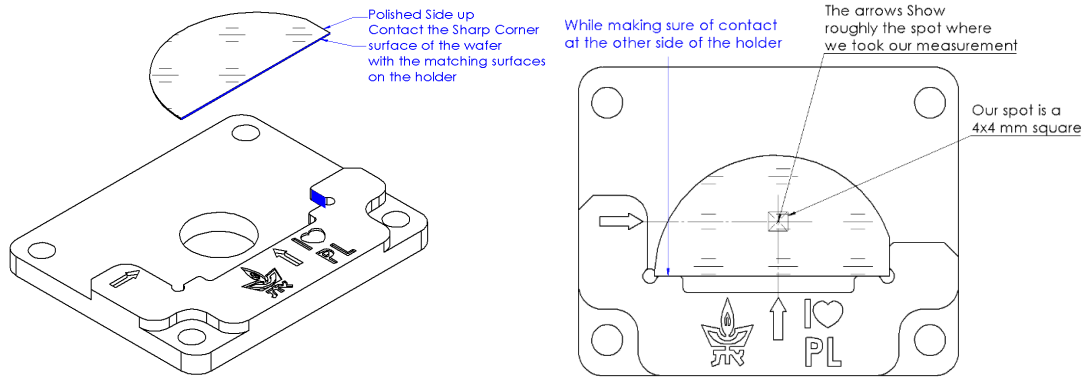


Figure G.1: 3D printed wafer holder with two adjacent alignment faces (bottom and side) assembly manual. This holder was used to ensure the position of the illumination is accurate for each measurement.

Integrating Sphere sample holders

The sample holders were designed to be positioned on the integrating sphere port. For reflectance PL measurement, a sample holder was designed to provide a large aperture and accurate and repeatable positioning on the sample port of the sphere, as shown in figure G.2. For transmission PL measurements, a sample holder was designed with the PL emission angle in mind, to allow recording of the entire PL, as shown in figure G.3. Each sample holder was specially designed to be placed in the sphere with any sample between $1 \times 1 \text{ cm}^2$ and half of the 2" wafer. wafer due to concerns regarding different PL signal results from different locations on the wafer. All sample holders were 3D printed in PLA.

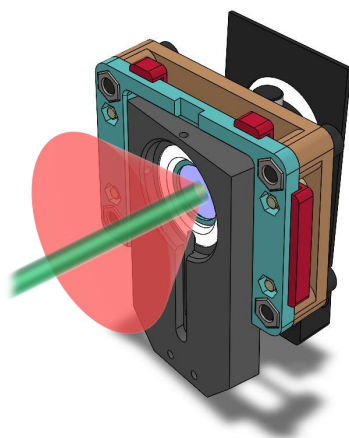


Figure G.2: 3D model schematics of the reflected PL wafer holder, showing the sample inside the holder (purple) under illumination (green beam). The holder is designed so that the sample is positioned flat and the PL emission (red) is not blocked and emitted into the integrating sphere.

FigureG.4 shows the model schematics of the measurement setups. FiguresG.4 a-b show the position of the sample holder (shown in figure G.2 in the integration sphere setup or the free space setup. The free space setup also includes x-y-z stages to allow for accurate positioning of the sample. This setup also includes a kinematic stage that allows to change between a vertical sample and a horizontal sample position, as shown. Vertical samples correspond to samples that require up-right positioning, such as electrochemical cells with liquid.

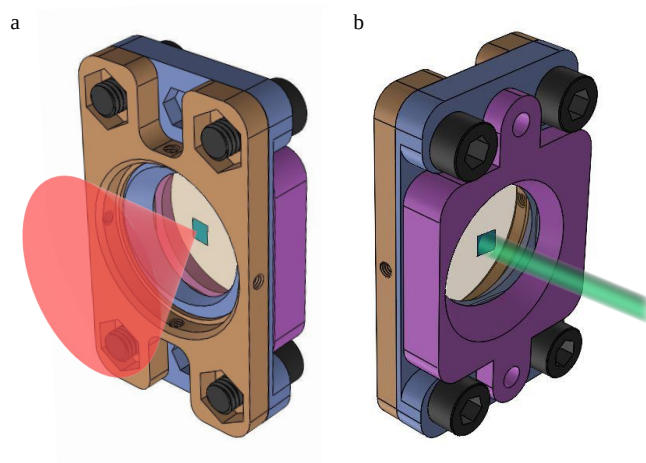


Figure G.3: 3D model schematics of the transmitted PL wafer holder, showing the sample inside the holder (semi-transparent) under illumination (green beam). The right view showing the illumination side and the left view shows the back side of the holder that is connected to the integrating sphere. The small square indicating the beam size on the sample. The holder is designed so that the sample is positioned flat and the PL emission (red) is not blocked and emitted into the integrating sphere.

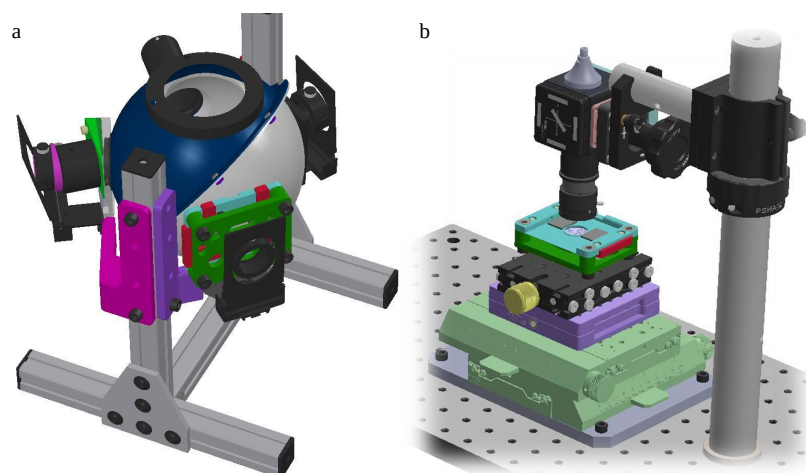


Figure G.4: 3D model schematics of the measurement setups built in SolidWorks. (a) The integrating sphere setup, showing the sample holder on the sample port as well as the power meter on a different port of the sphere to prevent substitution errors. Each holder lays on a 3D printed positioner attached to the sphere's frame (in purple and pink). (b) The free space setup, showing the sample holder facing the incident illumination from the top. This setup includes an x-y-z stages to allow for accurately positioning of the sample, as well as a kinematic stage to change between horizontal and vertical measurements to allow for vertical samples such as electrochemical cells.

MOS capacitor sample holder

In this section, the sample holder for electrical characterization of MOS capacitor structures. Initial measurements used conductive foam as the contact medium. However, this approach yielded inconsistent results, probably because of the non-uniform pressure distribution across the layered structure. Mechanical stress induced by the foam may have also introduced artifacts in capacitance-voltage measurements, particularly affecting hysteresis behavior. For this reason, sample holders that utilize spring-loaded "pogo" pins were used as electrical contacts. This design offers minimal mechanical distortion of the sample and reliable electrical connection. The SPCB-1 spring clip-board (Bridge Technology) was used in these measurements, which also helped with a fast sample change. The holder design incorporates separate pins for signal and ground connections, reducing parasitic capacitance effects. The sample holder is shown in Figure G.5.

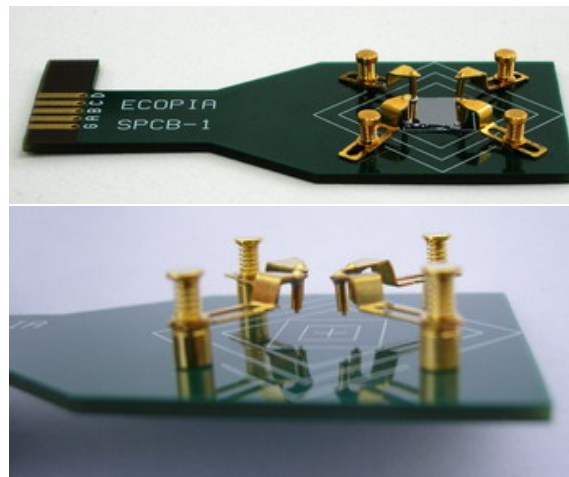


Figure G.5: MOS capacitor sample holder. The top picture shows the holder with a sample inside. The small spring loaded contacts (showed in the bottom picture) were used to gently connect the upper gate of the capacitor, and a conducting rubber was placed at the bottom to connect the bottom contact. gold plated spring the MOS capacitor experiments.

G.2.2 Photo-diode holder

A photo-diode holder was designed for a redundant power measurement setup alongside the commercial NEWPORT power-meter. This custom holder ensured precise positioning and stable operation of the photo-diode, enabling cross-checks and improved confidence in the measured optical power. The photo-diode was secured into the case and its contacts welded to a BNC output, securing its location with similar distance to the interface with the integrating sphere surface, as exists in the power meter.

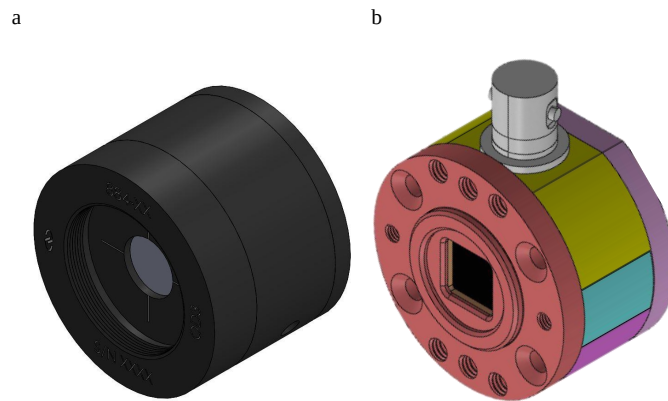


Figure G.6: A comparison between the 818-UV/DB Newport photodetector (left) and the design for the S175C photodetector holder. The holder was designed so that the detector is at the same distance from the external housing as in the Newport detector housing, in order to ensure the same measurement conditions and beam size on the detector.

תקציר

נוכח משבר האקלים, ולאור העלייה הצפויה בביקוש לחשמל כתוצאה מגלובליזציה, הרחבת הנגישות לאנרגיה ומהגידול באוכלוסיית העולם, הצורך בפתרונות אנרגיה מתחדשת קריטי. בשנים האחרונות תאים פוטו-וולטאיים (תאים סולאריים\תאי שמש) התגבשו כמועמדים המובילים להתמודדות עם אתגר זה.

טכנולוגיית התאים הסולאריים כיום לרוב מבוססת על תאי צומת יחיד מסיליקון, אשר מציעה המרת אנרגיה אמינה של קרינת השמש לאנרגיה חשמלית. התקנים אלו הגיעו לייעילות המרת אנרגיה המתקרבת בצורה מרשימה לגבול התרמודינמי האפשרי מהתקנים מסוג זה. בנוסף, תאי שמש מבוססי סיליקון הפכו לאחרונה לשיטה המשתלמת ביותר לייצור אנרגיה, תוך שהם עוקפים גם מקורות קונבנציונליים וגם מקורות אנרגיה מתחדשים אחרים. עם זאת, כל שיפור בטכנולוגיות המרת האנרגיה וביעילות של תאי השמש יוכל להיות מתורגם ישירות לצמצום השטח הנדרש לייצור אנרגיה, להגדלת היקפי הייצור וההפצה, ולבסוף להקטנת עלויות וליישום רחב יותר. לפיכך, שיפור יעילות המרת האנרגיה של תאים סולאריים הוא קריטי להבטחת עתיד אנרגטי בר-קיימא.

יעילותם של תאי השמש נקבעת על ידי תכונות הולכת המטענים ומנגנוני איבוד אנרגיה בתוך ההתקן. הערכה כמותית של מנגנונים אלה היא יסוד לאופטימיזציה של התקנים ולבחינה של חומרים חדשים. עבודה זו מציגה שיטה למיפוי כמותי של מנגנוני אובדן והולכת מטענים בעומק התא. באמצעות שילוב מדידות פוטולומינסנציה (PL) יחד עם מידול אופטי, עבודה זו מדגימה שיטה לחילוף של פרופיל יעילות הלומינסנציה המרחבית (SELE) כתלות בעומק. פרופיל ה-SELE מוגדר כהסתברות שזוג אלקטרון-חור אשר נוצר בעומק מסוים בתוך ההתקן יתרום לפליטת PL מההתקן. לפיכך, המיפוי מתאר תחרות בין רקומבינציה רדיאטיבית ורקומבינציה לא-רדיאטיבית בתוך ההתקן, ויכול לשמש לחקר תכונות של תכונות חומרים, ממשקי ופני שטח. מאחר שהיעילות הלומינסנציה קשורה למתח במעגל פתוח של ההתקן, ה-SELE ניתן לשימוש גם למיפוי התרומה המרחבית למתח של ההתקן.

שיטת חילוף ה-SELE מודגמת באמצעות מדידות PL של וויפרים עשויים GaAs וכן מסימולציות COMSOL. מוצג פיתוח של סימולציות PL וכן מודל של פליטה ובליעה של פוטונים בתא (Photon recycling). ה-SELE המחולץ ממדידות PL מיושם במקביל לסימולציות של SELE אשר מאשרות את הממצאים והשיטה ומאפשרות ייחוס מאפיינים בפרופיל ה-SELE לגדלים פיזיקליים חשובים כגון מהירות הרקומבינציה המשטחית וזמן חיים של נשאי המטען. בהמשך, מוצג ה-SELE של צמתי PN, אשר מדגים את פרופיל ה-SELE עבור התקנים מעשיים. לאחר מכן, באמצעות שימוש בקבל MOS ובשיטת ה-SELE, מודגמות תכונות פני שטח שונות הנראות בפרופיל ה-SELE. לבסוף מוצגות מדידות PL דינמיות וכן שינויים ב-SELE הנמדדים במהלך תהליך היווצרות שכבת תחמוצת טבעית על מצע GaAs.

אוניברסיטת תל אביב

הפקולטה להנדסה ע"ש איבי ואלדר פליישמן

בית הספר לתארים מתקדמים ע"ש זנדמן סליינר

מדידות וסימולציה של יעילות התרומה המרחבית לפוטו- לומינסנציה של פרוסות גליום-ארסניד

חיבור זה הוגש כעבודת מחקר לקראת התואר "דוקטור" בהנדסת חשמל

על ידי

תמיר ישורון

העבודה נעשתה בבית הספר להנדסת חשמל

בהנחיית ד"ר גדעון שגב

שבט תשפ"ו

אוניברסיטת תל אביב

הפקולטה להנדסה ע"ש איבי ואלדר פליישמן

בית הספר לתארים מתקדמים ע"ש זנדמן סליינר

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תמיר ישורון

שבט תשפ"ו